QUALIFICATION STANDARD AND PLAN

SONAR OP – RQ MS

SONAR Operator (Sensor Supervisor) 00324

NQUAL – ALBC

MITE ID – 123260

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
ORIGINAL		10 Mar 04	CHAP 1 & 2 (DMTE)
ORIGINAL – P Book		13 Oct 05	CHAP 3&4 (CFNOS STDS)
S-1		16 Nov 05	CHAP 1 & 2 (DMTE 3371- 4975-1 (DMTE 2-2-2/57800))
S-2		21 Nov 06	CHAP 1 & 2 (DMTE 3371-4798-1 (DMTE 2-2-2/91401)
P-1		21 Nov 06	CHAP 3 & 4 (DMTE 3371-4798-1 (DMTE 2-2-2/91401)
QSP-MS: 1.00		21 Nov 06	DMTE 2-2-2
1.1		06 Jun 11	CHAP 3 & 4 CFNOS
1.2		23 May 12	CFNOS
1.3 HCM Hybrid		2 May 13	CFNOS
1.4 HCM Compliant		11 July 13	CFNOS
1.5		20 Aug 2015	CNPTG
2.0	Major	31 Mar 2017	NTDC(A)
2.1	Major	30 Apr 2019	NPTG/NTDC(A)
2.2	Minor	04 Nov 2020	NTDC(A)
3.0	QSP Conversion	30 Jan 2023	NPTG

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ACRONYMS AND ABBREVIATIONS

A

ACCON	Acoustic Emissions Control
ACDR	Active Counter Detection Range
ADR	Active Detection Range
AF	Assessment Factor
AIT	Advanced Intelligence Tape
AOO	Area of Operations
AOP	Area of Probability
ARPS	Acoustic Range Prediction System
ARS	Acoustic Result Sheet
ASW	Anti-Submarine Warfare
ASWC	Anti-Submarine Warfare Coordinator
ATT	Actual Training Time

B

BAT	Base Administrative Time
BIT	Built-In Test

C

CAF	Canadian Armed Forces
CANTASS	Canadian Towed Array Surveillance System
CDR	Counter Detection Range
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
CPA	Closest Point Of Approach
CS	Command System
CSNI	Consolidated Secret Network Infrastructure
CTC	Contact

D

DAOD	Defence Administrative Orders and Directives
dB	Decibel
DNCM	Departmental Noise Control Monitor
DNCO	Departmental Noise Control Officer
DND	Department of National Defence
DNTE	Director Naval Training & Education
DPGR	Director Personnel Generation Requirements
DSCDS	Defence Subject Classification and Disposition System
DWAN	Defence Wide Area Network

E

EC	Enabling Check
EDI	Explain, Demonstrate and Imitate
EMATT	EMATT
EMCON	EMCON
EO	Enabling Objective

EQ	Experience Qualification
ESM	Electronic Sensor Measures
F	
FAB	Fixed Action Button
FC	Frequency Corrected
FO	Frequency Observed
FOM	Figure of Merit
FR	Frequency Received
G	
GCCS-M	Global Command and Control System - Maritime
GPNet	General Purpose Network
H	
HDDR	High Density Data Recorder
HMCS	Her Majesty's Canadian Ship
HOD	Head of Department
HYFIX	Hyperbolic Fixing
I	
IAW	In Accordance With
ID	Identification
IGOSS	International Global Oceanographic Survey System
IMS	Information Management Standard
IS	Information Systems
IT	Information Technology
IUSS	Integrated Underwater Surveillance System
J	
JEZEBEL	Low Frequency, Passive Acoustic Sonobuoy System
L	
LG	Lesson Guidance
LO	Learning Objective
LOFAR	Low Frequency Analyzing And Recording
LOFIX	Low Interest Fixing
LOS	Line Of Sound
M	
MCT	Main Cooling Tray
METOC	Meteorological And Oceanographic Center
MESIP	Military Employment Structure Implementation Plan
MILOC	Military Oceanography
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
MSDS	Material Safety Data Sheet
MVAR	Magnetic Variation
N	
NAVORD	Naval Order
NCM	Noise Control Monitor
NFS	Naval Fleet School

NFS(A)	Naval Fleet School (Atlantic)
NFS(P)	Naval Fleet School (Pacific)
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NQual	National Qualification
NTDC(A)	Naval Training Development Centre (Atlantic)
NTI	Naval Training Instruction
O	
OFA	Ocean Feature Analysis
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
OPDEF	Operational Deficiency
ORBAT	Order of Battle
OSN	Own Ship Noise
P	
PAL	Passive Aural Listening
PC	Performance Check
PCDR	Passive Counter Detection Range
PLAR	Prior Learning Assessment and Recognition
PO	Performance Objective
PRB	Progress Review Board
PSR	Predicted Sonar Range
PT	Physical Training
PTF	Proficiency Training Facility
PTP	Portable Target Programmer
Q	
QAB	Quick Action Button
QSP	Qualification Standard and Plan
QSPWB	Qualification & Standard Procedures Writing Board
R	
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
RCU	Remote Control Unit
RCVR	Receiver
REC/REP Sys	Recorder/Reproducer System
RELVEL	Relative Velocity
RF	Radio Frequency
RFI	Request for Information
RFS	Request for Support
ROW	Rest Of World
RPM	Rotation Per Minutes
RQ	Rank Qualification
RUC	Range Under The Curve
S	
SAT	Satisfactory

SEACOT	Sea Combat Operator Trainer
SETT	Shipboard Embedded Training Tool
SIAP	Sonar Interference Avoidance Plan
SBP 3-1	Sonobuoy Processor Ver. 3-1
SCS	Sonar Control Supervisor
SLC	Sonobuoy Launch Container
SO	Standing Order
SOA	Speed Of Advance
SONO	Sonobuoy
SOP	Standard Operating Procedure
SPS	Sonobuoy Processing System
SPF	Sonar Performance Figure
SRC	Speed Related Component
SRPM	Shaft Rotation Per Minute
SSD	Shipboard Standard Display
SSRS	Sonobuoy System Receiver Set
T	
TA	Training Authority
TAS	Towed Array Sonar
TAT	Training Administrative Time
TBP	Time Bearing Plot
TCG	Time Code Generator
TD	Temporary Duty
TDA	Torpedo Danger Area
TDA	Tactical Decision Aid
TE	Training Establishment
TEL	Technology Enabled Learning
TFP	Time Frequency Plot
TMA	Target Motion Analysis
TO	Training Officer
TPK	Turn Per Knot
TPO	Training Petty Officer
TRB	Training Review Board
TS	Training Site
TTT	Training Travel Time
U	
UNSAT	Unsatisfactory
UPS	Uninterruptible Power Supply
UWW	Under Water Warfare
UWWD	Under Water Warfare Director
V	
VELO	Velocity
VFK	Variable Function Keys
VGR	Video Graphic Recorder
Vol	Volume

W

WPN	Weapon
WHMIS	Workplace Hazardous Material Information System
WRT	With Regards to

X

XBT	Expendable Bathy Thermograph
XSV	Expendable Sound Velocity

FOREWORD

a. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: **http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page**.

b. A trainee will be granted the National Qualification (NQual) ALBC when the standards listed in all POs have been met.

c. A trainee will be granted the Military Individual Training and Education (MITE) identification (ID) 123260 when standards listed in all POs have been met.

d. The designated TEs for this training include all NFSs within NPTG.

TE	Phase / Module	MITE course ID	NQual
Naval Fleet Schools	Sensors Supervisor	123260	ALBC

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

a. The aim of this QSP is to establish the requirements and training for the SONAR OP – RQ MS (ALBC). This QSP is designed to assist NTDC(A) in developing, NFSs in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

b. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	119	
Enabling Checks (ECs)	35	
Performance Checks (PCs)	26	
ATT SUB TOTAL:	**180**	**22.5**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	2	
Training Administrative Time (TAT)	2	
Training Travel Time (TTT)	0	
Physical Fitness (PT)	15	
BAT, TAT, TTT, PT SUB TOTAL:	**19**	**2.375**
***Partial days are rounded up** **TOTAL:**	**190**	**25 days**

REQUIRED COURSE STAFFING

c. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Course Officer	Lt(N)	1	NWO 00207, ALBV	Level III Top Secret	All POs
Senior Instructor	PO1	1	Sonar Op 00324, ALBV	Level III Top Secret	All POs
Instructor	PO2	1	Sonar Op 00324, ALBD	Level II Secret	LG 001.01.03 LG 002.01.02

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Instructor	MS	1	Sonar Op 00324, ALBC	Level II Secret	All POs
CANTASS Operators	S3-S1	2	Sonar OP 00324, any of the following: ALFT,ALFU,AKYF	Level II Secret	PO 002
HMS Operator	S3-S1	1	Sonar OP 00324, any of the following: ALFT,ALFU,AKYF	Level II Secret	PO 003

CRITICAL TRAINING RESOURCES	# OF DAYS
CANTASS Mission Simulator (CMS) Trainers – PO 002	11
SEACOT – PO 001, PO 003	13
Classroom Classified	3
Classroom Unclassified	4

COURSE CAPACITY

d. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

a. Maximum – 12;

b. Minimum – 4; and

c. Optimum – 8.

PREREQUISITES

e. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Possess at minimum a Level II Security Clearance; and

b. Must have successfully completed:

(1) Acoustic Operator Qualification (Sonar Op – S1 (AKYF));

(2) Advanced Acoustic Analysis Course (AEJR); and

(3) IAW NPI 13 Annex C.

QUALIFICATION

f. Successful completion of AAAC (123261) and RQ-MS (123260) or SONAR Watch Leader (119844)-AKYG for submariners constitutes the primary means of attaining the RQ-Sonar Op-MS qualification (ALBC).

PRIOR LEARNING ASSESSMENT AND RECOGNITION

g. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

SPECIAL INSTRUCTIONS

h. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS

a. **Performance**: Conduct Individual/Team Training Events (G0050/G0065).

b. Conditions:

a. Given:

(1) SEACOT, SEAWOLF or CPF;
(2) DWAN/CSNI access; and
(3) References.

b. Denied: Nil; and

c. Environment:

(1) SEACOT, SEAWOLF or CPF.

c. **Standard**: The Acoustic Junior Supervisor shall Conduct Individual/Team Training Events (G0050/G0065) by:

a. Planning/Designing training exercise (G0010), to include:

(1) Identifying Capabilities and Limitations of Platforms (B0045); and
(2) Compiling ASW tactical picture (E0005).

b. Assisting in individual/team training events (G0045/G0060) by:

(1) Operating the simulator/trainer systems (G0095); and
(2) Inputting training scenarios (G0090).

d. **Training Limitations**: Limited Access Area.

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
LG 001.01.01 – INTERPRET INTELLIGENCE DATA

LG 001.01.01 – Interpret Intelligence Data		
Location Conditions	**Instructional Sequence**	
Classified Classroom	Beginning of course	
Time	**Classification**	**Instructor**
200 minutes	SECRET	RQ-MS Instructor
Learning Objectives	**Instructional Method(s)**	
1) Identify the organizations that provide fleet support to include: a. TRINITY INT, b. METOC; c. ADAC; d. Janes Publications e. DIAS Threat Web f. CFMWC; and g. FMF. 2) Interpret RFI/RFS's 3) Interpret Intelligence Data	1) T-Chart 2) Instructor led lecture 3) Student research 4) Flip Chart Presentation 5) RFI Confirmation Activity 6) Instructor led Consolidation Example	
Scenario	Nil.	
Activity Steps	1) Instructor will break the class into 2 groups, each group is given a flipchart or whiteboard. Instructor will pose question "What intelligence sources do we draw on for intelligence data?" Groups will write down all intelligence sources on their papers. Instructor will come around and check on the groups. Students will then share their lists with the other group, and instructor will discuss the Pros and Cons of each source and provide class with any missed answers. (Trinity, METOC, ADAC, CFMWC, Janes/DIAS) (10 minutes – Method 4) 2) Instructor to discuss the various units that provide intelligence data (LO 1) then lead students through navigating the various websites to include: a. TRINITY INT b. METOC c. ADAC d. CFMWC e. Janes/DIAS Threat Web\ (10 minutes – Method 2) 3) Students are then grouped (2-3 students per group) and assigned one of the websites to research and create a flip chart presentation on the information they would collect/gather using the assigned website. "<u>What information can be found on this website, and what is the significance?</u>" (30 minutes – Method 3)	

LG 001.01.01 – Interpret Intelligence Data

4) Students to then present their assigned topics to the class and Instructor to provide any information that was not covered by student presentation. Instructor to bring up the website after each group presents and demonstrate where to find info and its significance being specific to any information that was not presented by students. (Up to 10 min presentation, 10 min instructor feedback and content delivery per group)
(100 minutes – Method 1)

Anticipated Responses	
TRINITY INT	ORBAT How to determine base locations and fleet Foreign nation tactics
METOC	OFA Bottom Topography Potential Ranges
ADAC	Passive Acoustics
CFMWC	Canadian Doctrine Allied Doctrine References (ATP 28)
Janes/DIAS Threat Web	Platform Data Weapons Data

5) Instructor to discuss what an RFI is, what information is contained in RFIs and how to determine the relevant information.
(10 minutes – Method 2)

6) Each student provided an RFI and briefly write down/highlight what information they feel is relevant within that RFI. (4-5 key points).
(15 minutes – Method 5)
Students to then share their response and build a class list of the key information from the RFI provided. (5 minutes – Method 5)

7) Instructor to then conduct a lecture demonstrating for a given scenario (3 ships, 2 submarines, in a given AOO) what information you would need and reiterate where you would get that information IOT build a plan for a training exercise.
(20 minutes – Method 2)

LG 001.01.01 – Interpret Intelligence Data	
Instructor Resources	**References**
1) CSNI access 2) Flip Charts & Markers (x 5) 3) RFI example worksheet 4) Instructor scenario	1) NAVORD 2000-2 2) Websites: a. TRINITY INT; b. METOC; c. ADAC; d. CFMWC; and e. Janes f. DIAS Threat Web.
Note(s) to Instructor	
1) Instructor to introduce various supporting units by locating them on DWAN/CSNI. 2) The instructor will with the class interpret the provided data by identifying relevant information.	

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
LG 001.01.02 – IDENTIFY CAPS AND LIMS OF PLATFORMS

LG 001.01.02 – Identify Caps and Limitations of Platforms		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.01.01	
Time	**Classification**	**Instructor**
100 minutes	SECRET	RQ-MS Instructor
Learning Objectives	**Method(s)**	
1) Identify capabilities/limitations of friendly platforms 2) Identify capabilities/limitations of threat platforms	1) Student Research 2) Presentation	
Scenario	Nil.	
Activity Steps	1) Instructor to assign units for each student to research. Each student to be assigned: a. One friendly/ROW surface ship; b. One friendly/ROW A/C; and c. One threat/ROW submarine. (5 minutes – Method 1) 2) Instructor will show the example threat matrix to the students and then provide the template threat matrix for students to use in the next activity. 3) Students are then to use the previously instructed resources to conduct research on the assigned vessels and fill out the threat matrix template provided to include: a. Name/country; b. Emitters; c. Weapons (torpedoes, missiles & mines); d. Range/speed/run time of torpedoes; e. Method of homing; f. Speeds (max and best endurance); and g. Propulsion. (70 minutes – Method 2) 4) Students are to then share their information with the class indicating what sources they used. Instructor to allow opportunity to discuss if there is a conflict of information from differing sources. (25 minutes – Method 2)	
Instructor Resources	**References**	
1) CSNI workstation for each student 2) Threat matrix example 3) Threat matrix template 4) White board with markers	1) NAVORD 2000-2 2) Websites: a. TRINITY INT; METOC; ADAC; CFMWC; and Janes/DIAS Threat Web.	
Note(s) to Instructor		
1) Instructor will ensure students understand how to use the Threat Matrix template.		

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
LG 001.01.03 – COMPILE ASW TACTICAL PICTURE

LG 001.01.03 – Compile ASW Tactical Picture		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.01.02
Time	**Classification**	**Instructor**
150 minutes	SECRET	RQ-PO2 Instructor or above
Learning Objectives		**Method(s)**
1) Compiling ASW Tactical Picture 2) Understanding NTDS symbols		1) Student Reading 2) Best Summary 3) Jigsaw/Summary activity 4) Tabletop/ Instructor Led Lecture
Scenario	Nil.	
Activity Steps	1) Students will be broken up into groups 4 groups of 2-3 students and given one of the sections of ATP-28 Vol 1 Chapter 8 – ASWC Specified factors in ASW Defence of a Main Body. (5 min) a. 0801 - Introduction b. 0802 – Theoretical Factors in ASW defense c. 0803 – Limiting Lines of Approach d. 0806 – Application of Theoretical Principles (5 minutes – Method 1) 2) Each group will read and understand their given section and will create a Best Summary. Each group will then present their Best Summary to the class, with the instructor having input as required to ensure all topics of the sections are covered. (45 minutes – Method 2) 3) Instructor to lead an interactive lecture/tabletop of the following content to ensure all pertinent points are covered. Instructor to describe how this information is used in preparing the plan for exercise scenarios: a. Torpedo Danger Zone (TDZ); b. Limiting Lines of Submerged Approach (LLSA); c. Limiting Lines of Quiet Approach (LLQA); d. Point Contacts vs. Line of Bearing Contacts; e. Torpedo Danger Area (TDA); f. Datum g. Furthest On Circle (FOC) and TDA (expansion); h. Area of Probability (AOP, Circular and Rectangular); and i. Screen development (ship & helo sectors). (50 minutes – Method 4) 4) Students will review the MIL-STD-2525D Joint Military Symbology or visit the JIIFC webpage to become familiar with Symbology. Students will be provided a worksheet with various symbols, where they must determine what the symbol is for, or with a description, determine what the symbol looks like.	

LG 001.01.03 – Compile ASW Tactical Picture	
(50 minutes – Method 3)	
Instructor Resources	**References**
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard & markers 4) ASW Picture Compilation PowerPoint Lecture 5) NTDS Symbology Worksheet	1) ATP-1 2) ATP-28 3) CFCD 106 4) MIL-STD-2525D Joint Military Symbology 5) JIIFC COP MIL2525 Symbology webpage
Note(s) to Instructor	
1) Instructor may use either the MIL-STD-2525D Joint Military Symbology publication or the JIIFC COP MIL 2525 Symbology webpage if students have CSNI access. 2) JIIFC COP MIL 2525 CSNI webpage: (osint-jiifc.cmil.ca/geo_maps/COP_Team_Website/Google_Earth/geo-data_published/symbols_Webpage/symbols_home.html)	

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
LG 001.01.04 – UTILIZING TACTICAL VOICE PROCEDURES

LG 001.01.04 – Utilizing Tactical Voice Procedures		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.01.03	
Time	**Classification**	**Instructor**
50 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives	**Method(s)**	
1) Utilizing Brevity Codewords	1) Instructor Led 2) Student research 3) Student presentations	
Scenario	Nil.	
Activity Steps	1) Instructor to introduce APP 7 and how to use the publication. (5 minutes – Method 1) 2) Instructor to provide a list of 3-5 codewords per student to research within APP 7. Students are to determine what each codeword means and in what situation you would anticipate hearing it. (15 minutes – Method 2) 3) Students are then to present their codewords to the class. (25 minutes – Method 3) 4) Instructor will ensure the student's answers to the codewords are correct and answer any students' questions on various codewords meaning. (5 minutes – Method 1)	
Instructor Resources	**References**	
1) DWAN/CSNI workstation 2) Codewords List 3) Flipcharts/Whiteboard & Markers	1) APP 7	
Note(s) to Instructor		
1) Adjust presented codewords based on class size. If a large class have each student present ONE codeword, however for a smaller class size assign 2 (or up to 5) codewords.		

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
LG 001.01.05 – PLAN A TRAINING EXERCISE

LG 001.01.05 – Plan a Training Exercise		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.01.04
Time	**Classification**	**Instructor**
150 minutes	SECRET	RQ MS Instructor
Learning Objectives		**Method(s)**
1) Plan Training Exercises		1) Instructor led lecture 2) T-Chart 3) Presentation 4) Collaborative exercise
Scenario	Nil.	
Activity Steps	1) Instructor to provide students with a copy of a scenario script and explain the layout and the content that is included. (10 minutes – Method 1) 2) Students are then to break into 2 groups and create a T-Chart to break down the scenario. (25 minutes – Method 2)	

Scenario Requirement	Action/Outcome
- Riser detection - Torpedo firing	- Submarine surfaced/PD (set submarine depth to 0) - Fire torpedo from threat submarine

	3) Students are then to present their T-Chart to the class with any feedback on student selection provided by instructor. (15 min) (15 minutes – Method 3) 4) Students are then put into pairs and provided a “scenario criteria” list. Students are to work together to draft a scenario script. (60 minutes – Method 4) 5) Upon completion, students are to present their scenario. Students may use whiteboard to ‘tabletop” the scenario script/explain their script. (35 min) (35 minutes – Method 3) 6) Instructor to provide feedback to students on their developed scripts. (5 min) (5 minutes – Method 1)
Instructor Resources	**References**
1) DWAN/CSNI workstation 2) Flip chart/whiteboard with markers	1) NAVORD 2000-2 2) APP 7

LG 001.01.05 – Plan a Training Exercise	
3) Example scenario 4) Scenario criteria 5) Scenario planning script template	3) ATP-1 4) ATP-28 5) CFCD 106 6) Websites: a. TRINITY INT; b. METOC; c. ADAC; d. CFMWC; and e. Janes f. DIAS Threat Web.
Note(s) to Instructor	
1) Instructor may create their own scenario if time permits.	

EO 001.01 – PLAN/DESIGN TRAINING EXERCISE
EC 001.01 – PLAN/DESIGN TRAINING EXERCISE

EC 001.01 – Plan/Design Training Exercise		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.01	Classified Classroom	After LG 001.01.05
Time	**Classification**	**Instructor**
250 minutes	SECRET	RQ MS Instructor
Scenario	IAW training exercise scenario.	
Assessment Steps	**Input** – Students will be provided with a scenario, a Request For Information (RFI) response, a scenario planning script template, and a Tactical Decision Aids questionnaire. Students may use references for this EC. **Performance** – Using the RFI and scenario, the students will identify all Threat, Friendly and Neutral forces provided in the scenario. Using the scenario planning script template, students will develop a training scenario to include the criteria IAW EC 001.01 checklist. Students will then label the Tactical Decision Aids on the questionnaire. (70 minutes) Students will then present their developed training scenario to the class and instructor. (180 minutes total, 15 minutes per student includes debrief) **Output** – Trainee must attain 18 / 23 SATs to pass EC IAW EC 001.01 checklist. **Feedback** – Student is provided written feedback via the EC 001.01 checklist and verbally by the instructor at the conclusion of student presentation.	
Instructor Resources		
1) Training exercise scenario 2) RFI response 3) Scenario planning script template 4) Tactical Decision Aids questionnaire 5) References		
Note(s) to Instructor		
1) EC 001.01 Practical assessment is designed to evaluate the trainee's ability to interpret intelligence information, plan a training exercise using the intelligence information, and identify the ASW Tactical Decision Aids. 2) Pass mark will be 18 / 23 SATs to pass EC IAW EC 001.01 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended.	

PROTECTED B (when complete)

EC 001.01 PLAN/DESIGN TRAINING EXERCISE			
Item	SAT	UNSAT	Remarks
1) Interpret the information from the RFI to identify the following:			
a. All Threats	☐	☐	
b. Friendly forces	☐	☐	
c. Neutral assets	☐	☐	
d. Mission	☐	☐	
2) Select platforms IAW scenario criteria:			
a. Friendly Ship	☐	☐	
b. Friendly Aircraft	☐	☐	
c. Threat Submarine	☐	☐	
3) Build scenario timeline based on RFI provided, to include following criteria:			
a. Timings	☐	☐	
b. Submarine based on: • Speed; • Range; and • Weapon(s).	☐	☐	
c. Surface combatant based on; • Speed; • Range; and • Weapon(s).	☐	☐	
d. Aircraft	☐	☐	
e. HVU/Merchant	☐	☐	
f. Biological	☐	☐	
g. Gain Contact	☐	☐	
h. Lost Contact	☐	☐	
i. Fire Weapon	☐	☐	
4) Labelled the following Tactical Decision Aids on the provided template to include the following points:			
a. TDZ (S)	☐	☐	
b. TDA (S)	☐	☐	
c. LLSUA	☐	☐	
d. LLQUA	☐	☐	
e. AOP (S)	☐	☐	
f. DATUM (S)	☐	☐	
g. POINT CONTACTS	☐	☐	
Score: /23			

EC 001.01 PLAN/DESIGN TRAINING EXERCISE		
Trainee must achieve 18 / 23 SATs on non-critical factors to pass this EC.		
Positive Feedback	Improvement Feedback	
Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
LG 001.02.01 – SET UP SIMULATOR/TRAINER SYSTEM

LG 001.02.01 – Set up Simulator/Trainer System		
Location Conditions		**Instructional Sequence**
SEACOT		After EC 001.01
Time	**Classification**	**Instructor**
400 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Learning Objectives		**Instructional Method(s)**
1) Operating SETT / SAFSIM by: a. Identifying Simulator/ Trainer uses b. Initializing SETT c. Determining SETT Parameters d. Implementing Tactical Environmental Functions e. Conducting Own Ship Functions f. Setting up control functions of 2D/3D Radar/LINK; and g. Acoustic Warfare Functions		1) Lecture 2) EDI
Scenario	Nil.	
Activity Steps	1) Instructor to present PowerPoint lesson to students on the Simulator/Trainer system overview to include: a. Capabilities and limitations b. Training roles c. Scenario management (pre-built vs custom) d. Exercise configuration e. SETT trainer locations f. SAF SIM specific details On completion, Instructor to conduct PowerPoint lesson on Initializing SETT, providing students a copy of the Performance Support (training aid). (50 minutes – Method 1) 2) Instructor to then explain and demonstrate the following components of the simulator/trainer system. Explain then demonstrate each point and have students use the system themselves to practice (while instructor conducts another walk-through demo) a. Determine SETT Parameters i. Environmental Features (sea state, wind, precipitation); ii. Unload/update/pause-resume; and iii. MFW configuration. b. Own ship functions i. Helm operator; ii. Own ship positioning; iii. Rudder angle panel; and	

LG 001.02.01 – Set up Simulator/Trainer System	
	iv. Doppler speed log. c. Acoustic Warfare Functions to launch/employ following systems within the training system: i. Sonobuoy launcher; ii. CANTASS functions; iii. HMS functions; iv. SPS functions; v. Torpedo control functions; vi. Missile control functions; vii. NIXIE functions; and viii. Bathy functions. d. Set up and Control functions of 2D/3D radar/LINK (simulation mode) e. Communication system Setup. (350 minutes – Method 2)
Instructor Resources	**References**
1) Lesson notes for Simulator/Trainer information 2) Job Aids/Performance Support	1) SEACOT Operating Manual
Note(s) to Instructor	
1) Instructor SAFSIM to be displayed on projector to explain and demonstrate the steps as listed in the LO. 2) Students to imitate each step after the explanation and demo of each portion of the lesson. Students may use the job aids/performance supports.	

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
LG 001.02.02 – CREATE GAME PIECES

LG 001.02.02 – Create Game Pieces		
Location Conditions	**Instructional Sequence**	
SEACOT	After LG 001.02.01	
Time	**Classification**	**Instructor**
150 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Learning Objectives	**Method(s)**	
1) Creating Game Pieces 2) Editing Game Pieces	1) EDI 2) Practical	
Scenario	Nil.	
Activity Steps	1) Instructor to explain how to create a game piece, and then demo it using the instructor console. After each explanation and demonstration, students to practice creating the game piece specified. Games pieces to be instructed to include: a. Create new surface (using absolute); b. Create new surface relative to game piece; c. Create new fixed wing game piece using absolute; d. Create new rotary wing; e. Create a new sub-surface; f. Edit game piece; g. Edit game piece IFF; and h. Edit game piece radar. (50 minutes – Method 1) 2) Students are to then practice creating game pieces. (100 minutes – Method 2)	
Instructor Resources	**References**	
1) Performance Support 2) Instructor console projector	1) SEACOT Operating Manual	
Note(s) to Instructor		
1) Only select game pieces have full functionality within the system. 2) Select weapons system first, to ensure game piece selected is accurate to scenario needs.		

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
LG 001.02.03 – CREATE/INPUT SCENARIO

LG 001.02.03 – Create/Input Scenario		
Location Conditions	**Instructional Sequence**	
SEACOT	After LG 001.02.02	
Time	**Classification**	**Instructor**
150 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Learning Objectives	**Method(s)**	
1) Load SAF SIM in prep mode 2) Create an Environment 3) Load an Environment 4) Create Scenario 5) Select Game Pieces 6) Input Scenario	1) EDI 2) Practical	
Scenario	Nil.	
Activity Steps	1) Instructor to explain how to: a. Load SAF SIM in prep mode; b. create an environment; c. load an environment; d. select game pieces; e. create scenario; and f. input a scenario. Instructor to then demo each action using the instructor console. After each explanation and demonstration, students to practice the specified action. (50 minutes – Method 1) 2) Students are to then practice creating environments and scenarios (100 min) based on the scenario criteria worksheets. (100 minutes – Method 2)	
Instructor Resources	**References**	
1) Performance Support 2) Scenario Criteria worksheets (x12)	2) SEACOT Operating Manual	
Note(s) to Instructor		
1) Nil.		

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
LG 001.02.04 – OPERATE SIMULATOR/TRAINER SYSTEM

LG 001.02.04 – Operate Simulator/Trainer System		
Location Conditions	**Instructional Sequence**	
SEACOT	After LG 001.02.03	
Time	**Classification**	**Instructor**
200 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Learning Objectives	**Method(s)**	
1) Run a Training Scenario	1) EDI 2) Practical	
Scenario	Nil.	
Activity Steps	1) Instructor to explain how to run a training scenario, and then demo it using the instructor console. After each explanation and demonstration, students to practice the task specified. Instruction to include: a. Exercise setup; b. Assign Roles; c. Start; d. Manipulating Game pieces: i. alter kinematics (speed, altitude, direction, depth); ii. fire weapons; iii. activate jamming; and iv. activate IFF/LINK. e. Engagement override; f. Fault injection; and g. Tagging events for playback and debrief. (100 minutes – Method 1) 2) Students are then provided scenario criteria worksheets, where they will practice the skills of running a training scenario. (100 minutes – Method 2)	
Instructor Resources	**References**	
1) Performance Support 2) Scenario Criteria worksheet	1) SEACOT Operating Manual	
Note(s) to Instructor		
1) *IFF setup is to be taught but activating the IFF Interrogator during scenario can cause issues to SAFSIM if turned on. (This may change with future versions of SAFSIM).		

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
LG 001.02.05 – INPUT TRAINING SCENARIOS

LG 001.02.05 – Input Training Scenarios		
Location Conditions	**Instructional Sequence**	
SEACOT	After LG 001.02.04	
Time	**Classification**	**Instructor**
1200 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Learning Objectives	**Method(s)**	
1) Implement Training Scenarios	1) EDI 2) Practical 3) Scenario Demonstration	
Scenario	Nil.	
Activity Steps	1) Students are provided the training scenario they created in EC 001.01. The Instructor will explain and demonstrate how to enter the created scenario into SETT/SAFSIM using the following process, students will imitate: a. Set up the simulator/trainer system; b. Create game pieces; c. Create a scenario; and d. Run the simulator/trainer. (400 minutes – Method 1) 2) Students will be provided Training Scenario #1 criteria worksheet outlining the exercise requirements and an RFI response detailing the unit data. They are to plan/design a training event/scenario based on the requirements filing out the scenario script details. (60 minutes – Method 2) 3) Instructor to provide students feedback on the created scenario. (60 minutes, 5 mins x student) 4) Students will then enter Training Scenario #1 criteria into SETT/SAFSIM ensuring the conduct the following process: a. Set up the simulator/trainer system; b. Create game pieces; and c. Create a scenario. (100 minutes – Method 1) 5) Instructor will have the students run the simulator with Training Scenario #1 and use the PC 002 checklist as confirmation and to provide feedback. (180 minutes total, 15 min per student – Method 3) 6) Students will be provided Training Scenario #2 criteria worksheet outlining the exercise requirements and an RFI response detailing the unit data. They are to	

LG 001.02.05 – Input Training Scenarios	
	plan/design a training event/scenario based on the requirements filing out the scenario script details. (60 minutes – Method 2) 7) Instructor to provide students feedback on the created scenario. (60, 5 mins x student minutes) 8) Students will then enter Training Scenario #2 criteria into SETT/SAFSIM ensuring the conduct the following process: a. Set up the simulator/trainer system; b. Create game pieces; and c. Create a scenario. (100 minutes – Method 1) 9) Instructor will have the students run the simulator with Training Scenario #2 and use the PC 001 checklist as confirmation and to provide feedback. (180 minutes total, 15 min per student – Method 3)
Instructor Resources	**References**
1) Performance Support 2) EC 001.01 Scenarios 3) Training Scenario #1 4) Training Scenario #2 5) EC 001.02. Checklist 6) PC 001 Checklist	1) SEACOT Operating Manual
Note(s) to Instructor	
1) Nil	

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
EC 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS

EC 001.02 – Assist in Individual/Team Training Events		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.02	SEACOT	After LG 001.02.05
Time	**Classification**	**Instructor**
350 minutes	SECRET	RQ MS Instructor NCIOP RQ MS Instructor SONAR
Scenario	IAW training exercise scenario.	
Assessment Steps	**Input** – Each student is provided their own SETT/SAFSIM simulator console and will be provided with a Training exercise scenario. **Performance** –Using the Training exercise scenario, the students will: a. Set up the SETT/SAFSIM simulator/trainer system; b. Create game pieces; and c. Create a scenario. (110 minutes) Students will then run the SETT/SAFSIM simulator/trainer with the scenario they have entered and show it to the instructor for evaluation. (180 minutes total, 15 minutes per student) **Output** – Trainee must attain 28 / 35 SATs to pass EC IAW EC 001.02 checklist. **Feedback** – Student is provided written feedback via the EC 001.02 checklist and verbally by the instructor during student presentation to the class. (60 minutes total, 5 minutes per student)	
Instructor Resources		
1) EC Training exercise scenario 2) References		
Note(s) to Instructor		
1) EC 001.02 Practical assessment is designed to evaluate the trainee's ability to input a training scenario into SETT/SAFSIM 2) Pass mark will be 28 / 35 SATs to pass EC IAW EC 001.02 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended.	

PROTECTED B (when complete)

EC 001.02 ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS			
Item	SAT	UNSAT	Remarks
1) Setup Simulator/Training System IAW scenario requirements to include the following:			
a. Display Setup	☐	☐	
b. Communications	☐	☐	
c. NIXIE	☐	☐	
d. SENSORS	☐	☐	
2) Create a Scenario to include the following Game Pieces:			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/Merchant	☐	☐	
e. Biological	☐	☐	
3) Input the following Game Pieces:			
a. Submarine Game Piece	☐	☐	
b. Surface Combatant Game Piece	☐	☐	
c. Aircraft Game Piece	☐	☐	
d. HVU/Merchant Game Piece	☐	☐	
e. Biological	☐	☐	
4) Modify each created Game Piece to initial operating parameters outline in your Scenario:			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/Merchant	☐	☐	
5) Input Range Rings and attach to Game Piece(s) to indicate the following:			
a. Torpedo Max Range	☐	☐	
b. CANTASS Detection Range	☐	☐	
c. HMS Detection Range	☐	☐	
d. SPS Detection Range	☐	☐	
6) Start Scenario:			
a. Conduct Communications check(s)	☐	☐	
b. Give Surface Sitrep	☐	☐	
7) Conduct Game Piece alterations to the following pieces:			

EC 001.02 ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/Merchant	☐	☐	
e. Biological	☐	☐	
8) Conduct a Torpedo/Missile firing:			
a. Setup Torpedo	☐	☐	
b. Torpedo Launched	☐	☐	
c. Setup a Missile	☐	☐	
d. Missile Launched	☐	☐	
9) Followed the timelines IAW training scenario			
a. Timings:	☐	☐	
b. Sequence of Events	☐	☐	

Score: /35

Trainee must achieve 28/35 SATs on non-critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 001.02 – ASSIST IN INDIVIDUAL/TEAM TRAINING EVENTS
PC 001 – CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS

PC 001 – Conduct Individual/Team Training Events		
Scope	**Location Conditions**	**Instructional Sequence**
PO 001	Classified Classroom and SEACOT	After EC 001.02
Time	**Classification**	**Instructor**
500 minutes	SECRET	RQ-MS Instructor
Scenario	IAW training exercise criteria.	
Assessment Steps	**Input** – Each student is provided their own SETT/SAFSIM simulator console. Students will be provided with training exercise criteria and an RFI response detailing the unit data. **Performance** – This PC will be conducted in 2 parts: **Part 1 (Classified Classroom):** Students are to plan/design a training event/scenario based on the required criteria filling out the scenario script details. (50 minutes) **Part 2 (SEACOT):** Using their training plan/design, the students will: 1) Set up the SETT/SAFSIM simulator/trainer system; 2) Create game pieces; and 3) Create a scenario. (250 minutes) Students will then run the SETT/SAFSIM simulator/trainer with the scenario they have entered and show it to the instructor for evaluation. (180 minutes total, 15 minutes per student) **Output** – Trainee must attain 30 / 42 SATs to pass PC IAW PC 001 checklist. **Feedback** – Student is provided written feedback via the PC 001 checklist and verbally by the instructor during student presentation to the class. (20 minutes)	
Instructor Resources		
1) PC Training exercise criteria 2) RFI response 3) Scenario planning script template 4) Tactical Decision Aids questionnaire 5) References		
Note(s) to Instructor		
1) PC 001 Practical assessment is designed to evaluate the trainee's ability to plan/design a training scenario and input the training scenario into SETT/SAFSIM 2) Pass mark will be 30 / 42 SATs to pass PC IAW PC 001 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this PC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review	

PC 001 – Conduct Individual/Team Training Events	
	(IR) as a result of failure of this PC, continuation of training without a re-write may be recommended.

PROTECTED B (when complete)

PC 001 CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS			
Item	SAT	UNSAT	Remarks
1) Build scenario timeline based on RFI provided, to include following criteria;			
a. Timings	☐	☐	
b. Submarine based on; • Speed; • Range; and • Weapon(s).	☐	☐	
c. Surface combatant based on; • Speed; • Range; and • Weapon(s).	☐	☐	
d. Aircraft	☐	☐	
e. HVU/Merchant	☐	☐	
f. Biological	☐	☐	
g. Gain contact	☐	☐	
h. Lost contact	☐	☐	
i. Fire weapon	☐	☐	
2) Setup Simulator/Training System to include the following (G0095):			
a. Display Setup	☐	☐	
b. Communications	☐	☐	
c. NIXIE	☐	☐	
d. SENSORS	☐	☐	
3) Create a Scenario to include the following Game Pieces (E0005):			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/TM	☐	☐	
e. Biological	☐	☐	
4) Input the following Game Pieces (G0090):			
a. Submarine Game Piece	☐	☐	
b. Surface Combatant Game Piece	☐	☐	
c. Aircraft Game Piece	☐	☐	
d. HVU/TM Game Piece	☐	☐	
e. Biological	☐	☐	
5) Modify each created Game Piece to initial operating parameters outline in your scenario:			

PC 001 CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/TM	☐	☐	
6) Input Range Rings and attach to Game Piece(s) to indicate the following:			
a. Torpedo Max Range	☐	☐	
b. CANTASS Detection Range	☐	☐	
c. HMS Detection Range	☐	☐	
d. SPS Detection range	☐	☐	
7) Start Scenario:			
a. Conduct communications check(s)	☐	☐	
b. Give surface Sitrep	☐	☐	
8) Conduct Game Piece alterations to the following pieces:			
a. Submarine	☐	☐	
b. Surface Combatant	☐	☐	
c. Aircraft	☐	☐	
d. HVU/TM	☐	☐	
e. Biological	☐	☐	
f. Submarine	☐	☐	
9) Fire a Torpedo:			
a. Torpedo Launched	☐	☐	
10) Fire a Missile:			
a. Missile Launched	☐	☐	
11) Utilized CODE WORDS IAW with APP-7	☐	☐	

Score: /42

Trainee must achieve 30 / 42 SATs on non-critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard

PC 001 CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS		
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – DIRECT TOWED ARRAY SYSTEM OPERATIONS

a. **Performance**: Direct Towed Array System Operations (C0155).

b. Conditions:

a. Given:

(1) Towed Array Sonar System (TASS) equipment;
(2) Passive acoustic contacts;
(3) Beam/bearing converter;
(4) Logs and records;
(5) Communication equipment; and
(6) References.

b. Denied: Nil; and

c. Environment:

(1) At sea;
(2) Day or night; and
(3) Under any weather condition.

c. **Standard**: The Acoustic Junior Supervisor shall Direct Towed Array System Operations (C0155), to include:

a. Establishing capabilities and limitations of towed array operations in operation areas (C0160) to include:

(1) Interpreting Oceanographic Data (F0045);
(2) Calculating array tow speeds, tilt and/or balancing (C0120); and
(3) Ensuring Tow array Sonar system is set up (C0127).

b. Amplifying track information (E0020) to include:

(1) Assessing the UWW threat information (D0050) by;
(a) Supervising the tracking of acoustic contacts (C0070) to include;
(b) Analyzing passive acoustic data (C0065);
(c) Analyzing TMA data (C0095); and
(d) Identifying changes in contact's readiness and tactical states (F0015).
(2) Contributing to intelligence gathering operations (B0030) to include:
(a) Recording acoustic intelligence data (F0010); and
(b) Annotating recording media (C0075).

d. **Training Limitations**: Limited Access Area.

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
LG 002.01.01 – UNDERSTANDING SAFETY CONSIDERATIONS & RESTRICTIONS

LG 002.01.01 – Safety Considerations and Restrictions		
Location Conditions		**Instructional Sequence**
Unclassified classroom		After LG 002.01.02
Time	**Classification**	**Instructor**
50 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives		**Instructional Method(s)**
1) Interpreting NAVORD 3137-1 2) Identify capabilities and limitations of the towed array 3) Understanding Safety considerations		1) Think, "Pair", Share/T-chart 2) Student reading 3) T-Chart
Scenario	Nil.	

Activity Steps

1) Instructor to pose to class "What Safety Considerations and Restrictions would you have to consider with Towed Array Operations?" Make a list individually. Students are broken into 3 even groups and share their lists.
(5 minutes – Method 1)

2) Students are assigned to read NAVORD 3137-1. (25 min)
(20 minutes – Method 2)

3) Groups are then able to amend their lists based on the readings.
(5 minutes – Method 1)

4) Instructor to assign each group a heading "towing restrictions, operational restrictions or procedures". Students are then to present their assigned topic lists and create one class list.

Anticipated Responses	
Towing Restrictions	Sea State Own Ships Speed Area of Operations Echo Sounder Operations UWON RAM Procedures NIXIE Operations
Operational Restrictions	Array Safety (Weather) Shallow Water Towing Procedures Stopping Procedures Cable Scope Array Depth Rescue Stations Procedures RAS Procedures
Procedures	Strumming Depth Safety Limitations

LG 002.01.01 – Safety Considerations and Restrictions		
		Surface Restrictions (Broaching/Grounding) Use of Cable Cutter ISOPAR M Safety and Precautions
	(15 minutes – Method 3) 5) Instructor to reinforce topics in wrap up to ensure students are familiar with the considerations regarding: a. Deployment and Retrieval Limitations; b. Speed Considerations; and c. Depth and Surface considerations. (5 minutes)	
Instructor Resources		**References**
1) DWAN workstations 2) NAVORD 3137-1 paper (if DWAN not available) x 13 3) Flip charts & markers		1) NAVORD 3137-1
Note(s) to Instructor		
1) Nil		

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
LG 002.01.02 – INTERPRET OCEANOGRAPHIC DATA

LG 002.01.02 – Interpret Oceanographic Data		
Location Conditions	**Instructional Sequence**	
Unclassified classroom	After PC 001	
Time	**Classification**	**Instructor**
200 minutes	UNCLASSIFIED	RQ-PO2 Instructor
Learning Objectives	**Method(s)**	
1) Interpret Oceanographic Data 2) Identify Tactical Environmental Factors that apply to towed array operations 3) Optimize TAS placement	1) PowerPoint 2) Collaborative List/ T-Chart 3) Collaborative Problem Solving	
Scenario	Nil.	
Activity Steps	1) Instructor to present PowerPoint lesson for tactical considerations of oceanographic data and environmental factors regarding TAS placement to include at a minimum: a. Ocean Features (OFA); b. Bottom Topography; c. Bottom Composition; d. Sound Paths; e. Tactical Implications of Cut-off Frequencies; and f. Ambient Noise Sources. Students are to be provided a copy of the PowerPoint lesson (training aid). (100 minutes – Method 1) 2) Students split into two teams, and they are assigned a topic to consider: advantages to own ships sensors or advantages for submarines operating for Towed Array Operations. (15 minutes – Method 2)	

Oceanographic/Environmental Factors	
Advantages to own ships sensors	Advantages for submarines operating
- Hard sandy bottom (Shallow water) can benefit sonar performance. - Use of Convergence Zone - Use of SLD for sensors placements. - Use of cold Eddies (Decreased range between CZ) - Use of megaphone effect	- Use of shadow zones caused by bottom topography. - Use of the SLD for navigation i.e., surface duct. - Use of high shipping area. - Use of Warm Eddies for navigation (Increased range between CZ)

LG 002.01.02 – Interpret Oceanographic Data	
3) Students are then to present their assigned topic to the class with any feedback on student selection provided by instructor IOT build a class T-Chart. (20 minutes – Method 2) 4) Students are then broken into teams of 2-3 and provided an oceanographic/environmental scenario they must break down and determine the critical tactical considerations regarding the scenario and optimal TAS placement. (15 minutes – Method 3) 5) Students are to then present their scenario to the class and discuss the tactical considerations with feedback and discussion amongst the class. (50 minutes – Method 3)	
Instructor Resources	**References**
1) DWAN workstation 2) Projector 3) Flipchart with markers 4) Oceanographic scenarios (x 4)	1) MILOC 2) RP33
Note(s) to Instructor	
1) Nil	

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
LG 002.01.03 – CALCULATE ARRAY TOW SPEEDS

LG 002.01.03 – Calculate Array Tow Speeds		
Location Conditions		**Instructional Sequence**
Unclassified classroom		After LG 002.01.01
Time	**Classification**	**Instructor**
100 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Calculate Array tow speeds, tilt & balancing 2) Estimate Array stabilization time		1) Student exploratory learning 2) Calculation scenario 3) Instructor led 4) Student presentations
Scenario	Nil.	
Activity Steps	1) Instructor to provide the table AND the manual calculations (cable scope/ships speed)/2. (10 minutes) 2) Students are then broken into pairs and provided a different scenario with missing data. They must use the table AND manually calculate to determine missing data. (40 minutes – Method 2) 3) Instructor to then demonstrate to students how to solve the problem based on a scenario and how to estimate array "at depth" and stabilization time for command. (5 minutes – Method 3) 4) Students are to then conduct any corrections, determine time required, and present their calculations regarding their scenario. (15 minutes – Method 2) 5) Students are then broken into new pairs and provided a new scenario where they must determine the missing data and provide impact to command. (10 minutes – Method 2) 6) Students are to then present their calculations and impact regarding their scenario. (20 minutes – Method 4)	
Instructor Resources		**References**
1) Calculation tables (NAVORD 3137-1) 2) Scenarios (x 14)		1) NAVORD 3137-1
Note(s) to Instructor		
1) Time considerations are critical to command and should be emphasized.		

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
LG 002.01.04 – ENSURING TOWED ARRAY SONAR SYSTEM SET-UP

LG 002.01.04 – Ensuring Towed Array SONAR System Set-Up		
Location Conditions		**Instructional Sequence**
CMS Lab (primary) SEACOT (secondary)		After LG 002.01.03
Time	**Classification**	**Instructor**
150 minutes	SECRET	RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Communications Set Up/Comm Check 2) Determine equipment requirements 3) Brief Team regarding Operational info 4) Ensure system parameters set-up 5) Adjust Magnetic Variation for AO 6) Delegate team duties and responsibilities 7) Conduct Closing Up Report IAW REF 8) Start/Stop tape recording		1) EDI
Scenario	Nil.	
Activity Steps	1) Instructor to explain the requirements as a supervisor for setting up the system. What you are specifically looking at/for, common operator errors, etc. (10 minutes – Method 1) 2) Instructor to then explain while demonstrating how to set up the system (15 min) (15 minutes – Method 1) 3) Students are then allocated a workstation and follow along with the instructor for system set up (as there may not be enough consoles per student, rotate through to ensure everyone has an opportunity to have a walk through with instructor). (125 minutes – Method 2)	
Instructor Resources		**References**
1) Error Scenarios (x 8)		2) NCPM-600
Note(s) to Instructor		
1) One Student to act as operator (with script while the other student acts as the supervisor).		

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
LG 002.01.05 – CLOSING UP THE WATCH

LG 002.01.05 – Closing up the Watch		
Location Conditions		**Instructional Sequence**
CMS Lab (primary) SEACOT (secondary)		After LG 002.01.04
Time	**Classification**	**Instructor**
250 minutes	SECRET	2x RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Assess UWW threat information 2) Ensure Tow Array Sonar System is setup 3) Identify Ownship changes and impacts to towed array operations		1) Practical
Scenario	Nil.	
Activity Steps	1) This lesson is a consolidation exercise to allow students an opportunity to develop skills prior to EC 002.01. 2) Students will conduct their duties as they would in "taking over the watch". They must set up the towed array sonar system, they are provided updated information regarding oceanography and threat, where they must then provide recommendations for sensor employment (including the time it will take to execute the change). Two students to conduct activity at a time, with student support as operators. a. System Set up and Closing Up report; 10 min/student and b. Review updated oceanography/threat data, calculate, and provide recommendation. 20 min/student (250 minutes – Method 1) 3) Students to be provided feedback IAW EC 002.01 Checklist	
Instructor Resources		**References**
1) Predicted ranges 2) CMS set up scenario 3) Oceanography data 4) Threat information/update		1) NAVORD 3137-1 2) CANTASS Operator Manual
Note(s) to Instructor		
1) Require two instructors IOT provide feedback to students.		

EO 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS
EC 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS

EC 002.01 – ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01	CANTASS Mission Simulator Trainer or SEACOT	After LG 002.01.05
Time	**Classification**	**Instructor**
400 minutes (30 Minutes per student)	SECRET	RQ MS Instructor
Scenario	IAW EC 002.01 scenario.	
Assessment Steps	**Input** – Students conduct this exercise in the CANTASS Mission Simulator, they will be provided with CANTASS console and 1-2 operators, a written scenario, calculator and NAVORD 3137-1 for reference. **Performance** – Students will be required to read the scenario, they will provide verbal and/or written answers to the instructor/assessor for Parts 1 – 4 of EC 002.01 Checklist. The instructor will then direct the CANTASS Operators to perform Part 5 of the EC 002.01 (360 minutes total, 30 minutes per student) **Output** – Trainee must attain 18 / 25 SATs to pass EC IAW EC 002.01 checklist. **Feedback** – Student is provided written feedback via the EC 002.01 checklist and verbally by the instructor upon completion of the EC (40 minutes)	
Instructor Resources		
1) CANTASS Mission Simulator 2) 2 x CANTASS Operators (1 minimum) 3) EC scenario 4) Calculator 5) NAVORD 3137-1 Tow Chart 6) EC 002.01 Checklist		
Note(s) to Instructor		
1) EC 002.01 Practical assessment is designed to evaluate the trainee's ability to establish the capabilities and limitations of the Towed Array during Operations. 2) Pass mark will be 18 / 25 SATs to pass EC IAW EC 002.01 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended.	

PROTECTED B (when complete)

EC 002.01 ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS			
Item	**SAT**	**UNSAT**	**Remarks**
1) Determine the safe towed operating depth(m) parameters IAW A6:			
a. Minimum operating depth	☐	☐	
b. Maximum operating depth	☐	☐	
c. Reporting frequency	☐	☐	
d. Broach depth reporting to OOW	☐	☐	
e. Indicate the maximum Sea State that towed array deployment and retrieval can occur	☐	☐	
2) Calculate the ships speed for array depths(m) of:			
a. Maximum safe depth	☐	☐	
b. Minimum safe depth	☐	☐	
c. Recommended towed array depth	☐	☐	
3) Calculate the towed array depth at the indicated Array Scope for the following speeds:			
a. 8 knots	☐	☐	
b. 10 knots	☐	☐	
c. 12 knots	☐	☐	
d. 26 knots	☐	☐	
4) Calculate the amount of Array Scope is required to raise and lower the Array Depth, at the indicated speed by:			
a. 50 meters	☐	☐	
b. 100 meters	☐	☐	
Set up Procedures:			
a. Conduct communications check with SCS and bridge	☐	☐	
b. Ensure all required equipment is available to the team to include: • Beam/brg converter wheel; • Headsets; • Logs; • Tape in HDDR; and • HDDR set to remote.	☐	☐	

EC 002.01 ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS			
c. Conduct team brief to include: • Exercise/area/situation; • Friendly forces; • TOI; • Environmental conditions; and • Pertinent target detail.	☐	☐	
d. Ensure sanitization completed	☐	☐	
e. Ensure integration time for the following processing: • Narrow Band (NB); • Broadband (BB); and • Demon (D).	☐	☐	
f. Ensure verniers are set up IAW team brief to include: • V1 and V2 center frequencies as priority; • V3 and V4 center freq. as secondary; and • Bandwidths for all four verniers as required.	☐	☐	
g. Ensure array gain is set to optimize resolution	☐	☐	
h. Ensure operator adjust Magnetic Variation for Area of Operation	☐	☐	
i. Delegate operators as "Fwd searchers, AFT searchers" and logger	☐	☐	
j. TAS SUP gives detailed closing up report to SCS IAW A5	☐	☐	
k. Ensure tape is recording	☐	☐	

Score: /25

Trainee must achieve 18 / 25 SATs on non-critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard

EC 002.01 ESTABLISH CAPABILITIES/LIMITATIONS OF TOWED ARRAY OPS		
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 002.02 – AMPLIFY TRACK INFORMATION
LG 002.02.01 – SUPERVISE TRACKING ACOUSTIC CONTACTS

LG 002.02.01 – Supervise Tracking Acoustic Contacts		
Location Conditions		**Instructional Sequence**
Classified classroom		After EC 002.01
Time	**Classification**	**Instructor**
100 minutes	SECRET	RQ-MS Instructor
Learning Objectives		**Instructional Method(s)**
1) Supervise Acoustic Contact Tracking 2) Contribute to RUP (recognized underwater picture) 3) Direct Operators IAW display priorities		1) Tabletop 2) Collaborative Problem Solving
Scenario	Nil.	
Activity Steps	1) Instructor to lead a tabletop/interactive discussion explaining key factors in supervising acoustic contact tracking (TEL request to develop video to support). Content to include: a. Forward and Aft display management based on intelligence/threat; and b. Voice reports/Recommendations to include sensor employment. (30 minutes – Method 1) 2) Students are broken into teams or 2-3 and provided a scenario where they must determine sensor employment recommendations IOT track acoustic contacts (including verniers and placement of sensors). (20 minutes – Method 2) 3) Students to present their scenario and their recommendations to the class, with class discussion and feedback. (50 minutes – Method 2)	
Instructor Resources		**References**
1) Tracking scenarios (x6)		1) CFCD 113 Vol 2 2) NAVORD 3137-1
Note(s) to Instructor		
1) Nil		

EO 002.02 – AMPLIFY TRACK INFORMATION
LG 002.02.02 – DEVELOP TMA

LG 002.02.02 – Develop TMA		
Location Conditions		**Instructional Sequence**
Unclassified classroom		After LG 002.02.01
Time	**Classification**	**Instructor**
200 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Apply TMA theory 2) Plot technique exposure 3) Develop and verify TMA data a. LOS; b. TMA relationships; and c. RELVEL.		1) Instructor led lesson 2) Student worksheets
Scenario	Nil.	
Activity Steps	1) Instructor to deliver a lesson on the theory of TMA and provide examples with explanation of the plotting techniques. (40 minutes – Method 1) 2) Instructor to demonstrate an example problem with the class on the whiteboard. (30 minutes – Method 1) 3) Students are provided worksheets (courseware) to complete. (120 minutes – Method 2) 4) Instructor to take up the worksheet and provide feedback. (10 minutes – Method 1)	
Instructor Resources		**References**
1) TMA lesson 2) Instructor example problem 3) TMA worksheets (X 12) 4) TMA scenarios 5) HMCS VENTURE RELVEL videos (https://vimeo.com/channels/1562795/videos)		1) NCPM 600 2) NOSTP 201 REL VEL Handbook
Note(s) to Instructor		
1) If students require more time to practice RELVEL, they are to conduct refresher outside classroom hours. 2) Instructor may develop their own TMA / RELVEL scenarios as required.		

EO 002.02 – AMPLIFY TRACK INFORMATION
LG 002.02.03 – ANALYZING TMA

LG 002.02.03 – Analyzing TMA		
Location Conditions		**Instructional Sequence**
CMS Lab (primary) SEACOT (secondary)		After LG 002.02.02
Time	**Classification**	**Instructor**
50 minutes	SECRET	RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Interpret changes in contact position (RELVEL) 2) Conduct TMA (amplifying) report		1) EDI 2) Practical
Scenario	Nil.	
Activity Steps	1) Using the CANTASS Mission Simulator or SEACOT, the instructor will demonstrate a TMA scenario on CANTASS. (15 minutes – Method 1) 2) Students will then be provided opportunity to practice a TMA scenario and provide amplifying reports to the instructor. (35 minutes – Method 2)	
Instructor Resources		**References**
1) Go Operator (support staff)		1) CMS manual
Note(s) to Instructor		
1) Multiple displays tracking during scenarios. 2) Instructors to manipulate CMS on behalf of the SCS. 3) Instructor to circulate the room and provide feedback. 4) Instructor may use students as Go Operator if no support staff available.		

EO 002.02 – AMPLIFY TRACK INFORMATION
LG 002.02.04 – RECORD/ANNOTATE ACOUSTIC INT DATA

LG 002.02.04 – Record/Annotate Acoustic Intelligence Data		
Location Conditions		**Instructional Sequence**
Classified Classroom		Instructed after LG 002.02.03
Time	**Classification**	**Instructor**
100 minutes	SECRET	RQ-MS Instructor
Learning Objectives		**Method(s)**
1) Quality Assurance of Logs Maintenance and data recording 2) Handling of data tapes		1) Reading Assignment 2) QA activity 3) T-Chart 4) Brainstorming
Scenario	Nil.	
Activity Steps	1) Instructor to review the format for Logs and data recording. (5 minutes) 2) Students are to read NCPM 600 Annex C on: ADAC Log Standards for CANTASS, and review the CANTASS Log Example and Log Template. (10 minutes – Method 1) 3) Instructor to break the students up into 2 groups. Each group is provided half of the whiteboard or a flip chart. Students are to create a T-chart with "What is important in logs" on one side, and "Why" on the other. Each group will be given a chance to present their T-Chart to the other group. (20 minutes – Method 4) 4) Students are provided "exercise logs" to QA as the supervisor. (15 minutes – Method 2) 5) Students will read National Defence Security Orders and Directives Chapter 6: Security of Information Standards, section Standard BL Moving of Sensitive Material to include: a. Safeguard in the Movement of Sensitive Material; b. Travel with Sensitive Information; c. Packaging Material for Transport or Transmittal. (30 minutes – Method 1) 6) Students will be broken into 2-3 groups and each group is given a different scenario. Students will then brainstorm proper methods of handling and delivering classified material and present to the class. (20minutes – Method 4)	
Instructor Resources		**References**
1) NCPM 600 Annex C to include:		1) NCPM 600

LG 002.02.04 – Record/Annotate Acoustic Intelligence Data	
a. ADAC Log Standards, b. CANTASS Log Example; and c. CANTASS Log template 2) NDSODs Chapter 6: Security of Information Standards 3) Whiteboard/Flipchart & Markers 4) Exercise logs with errors 5) Packaging Scenarios	2) NDSODs Chapter 6: Security of Information Standards
Note(s) to Instructor	
1) Instructor to ensure coverage of both logs and handling/transfer of data tapes to ADAC.	

EO 002.02 – AMPLIFY TRACK INFORMATION
LG 002.02.05 – DIRECT TOWED ARRAY OPERATIONS

LG 002.02.05 – Direct Towed Array Operations Consolidation		
Location Conditions	**Instructional Sequence**	
CANTASS Mission Simulator (Primary) or SEACOT(Secondary)	After LG 002.02.04	
Time	**Classification**	**Instructor**
1200 minutes	SECRET	RQ-MS Instructor
Learning Objectives	**Method(s)**	
1) Conduct role as TASSUP 2) Direct Towed Array Operations	1) EDI 2) Practical 3) Debrief	
Scenario	Nil.	
Activity Steps	1) Instructor will lead students in a demonstration of how to conduct the position of Towed Array Supervisor (TASSUP). (40 minutes – Method 1) 2) Students will then conduct a 30 minute run as the TASSUP where they will be provided the following opportunities: a. Be provided a submarine contact that alters at least once; b. Conduct a Situation, Assessment, Recommendation (SAR) report to the SCS; and c. Supervisor operators to ensure voice reports, tracking and contact logging is conducted. Instructor will use the EC 002.02 checklist to ensure students complete all tasks required and provide students feedback on their run. (360 minutes total, 30 minutes per student – Method 1) 3) Students will be provided a minimum of two 30 minute practice runs per student, conducting the duties of TASSUP as they would in “taking over the watch”. IAW PC 003 checklist, students must ensure the following: a. System Set up and Closing Up report; b. Review updated oceanography/threat data; c. Calculate Towed Array Depth/Speed requirements; d. Provide recommendations to the SCS/Instructor on sensor employment; e. Supervise operator(s) to include: i. Operating the CANTASS; ii. Conducting contact search and tracking procedures: iii. Detection and reporting of contacts; iv. Ensure contact logging; f. Utilize TMA sensor information to determine tactical situation; and g. Make SAR reports to the SCS/Instructor on contacts and tactical situation. Students to be provided feedback IAW PC 003 checklist upon completion of their development runs. (800 minutes total, 65 minutes per student – Method 2 and 3)	

LG 002.02.05 – Direct Towed Array Operations Consolidation	
Instructor Resources	**References**
1) Beam Bearing Wheel 2) EC 002.02/PC 002 Checklist 3) Voice Reports 4) Oceanography/threat data Cards	1) NCPM 600
Note(s) to Instructor	
1) Instructor may have two students conduct this activity at the same a time or have more than one instructor/assessor 2) Students may act as CANTASS operators during development runs.	

EO 002.02 – AMPLIFY TRACK INFORMATION
EC 002.02 – AMPLIFY TRACK INFORMATION

EC 002.02 – Amplify Track Information		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.02	CANTASS Mission Simulator Trainer or SEACOT	After LG 002.02.05
Time	**Classification**	**Instructor**
500 minutes (1 x 40 min period per student)	SECRET	RQ-MS Instructor

Scenario	IAW EC 002.02 scenario.
Assessment Steps	**Input** – Students conduct this EC in either the CANTASS Mission Simulator or SEACOT, they will be provided with CANTASS console and 1-2 operators, a written scenario with a Line Of Sound diagram template, calculator and a subsurface contact to track. **Performance** – Students will be required to read the scenario, supervise the CANTASS operator's in the search and initial contact procedures, conduct TMA on a target, and supervise Lost Contact procedures IAW EC 002.02 checklist. (480 minutes total, 40 minutes per student) **Output** – Trainee must attain 22 / 31 SATs to pass EC IAW EC 002.02 checklist. **Feedback** – Student is provided written feedback via the EC 002.02 checklist and verbally by the instructor upon completion of the EC. (20 minutes)

Instructor Resources

1) CANTASS Mission Simulator
2) 2 x CANTASS Operators (1 minimum)
3) EC scenario with Line of Sound diagram template
4) Calculator
5) NAVORD 3137-1 Tow Chart
6) EC 002.02 Checklist

Note(s) to Instructor

1) EC 002.02 Practical assessment is designed to evaluate the trainee's ability to supervise search, initial and lost contact procedures, conduct TMA and provide amplifying track information as the TASSUP.
2) Pass mark will be 22 / 31 SATs to pass EC IAW EC 002.02 checklist.
3) References are permitted.

Assessment Limitations	Nil.
Impact of Failure	Failure to achieve the standard on this EC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended.

PROTECTED B (when complete)

EC 002.02 AMPLIFY TRACK INFORMATION			
Item	SAT	UNSAT	Remarks
1) Supervise search, contact procedures:			
a. Ensure detailed NB search is done (SS upper TVM, TBB lower TVM)	☐	☐	
b. Ensure detailed BB search is done	☐	☐	
c. Upon gaining CTC assign operator as "Tracker" and/or "Searcher"	☐	☐	
d. Ensure searcher sets up to cover full search duties (SS Comb upper TVM and TBB for lower TVM)	☐	☐	
e. Ensure the operator create TGT for TOI	☐	☐	
f. Ensure the operator enters 1 OIT and 1 OUM NB components as directed by the invigilator	☐	☐	
g. Ensure the operator enters TGT source components	☐	☐	
h. Ensure the operator Searcher enters two BB OUM components on the related tracked TGT, assigning them to the target detail of the target	☐	☐	
i. Ensure the operator Check for audio on bearing of target	☐	☐	
j. Ensure the operator Generate a CMS330 contact detail for applicable target ensuring all data is in the CCD Contact Detail	☐	☐	
k. Give detailed Initial Contact report to SCS IAW A5	☐	☐	
l. Ensure searcher updates applicable CCD data area for CTC i.e., Class. / estimate speed / BB / Audio	☐	☐	
m. Ensure use of display format; TBB in BRG stabilized on relevant beam and manual input of OUM data	☐	☐	
n. Ensure logs are accurate i.e., OCS / array data / TGT data and events / environmental conditions / system events	☐	☐	
o. Conduct detailed amplification report IAW A5	☐	☐	

EC 002.02 AMPLIFY TRACK INFORMATION			
p. Ensure the operator resolve for ambiguity: • Correct beams are utilized in proper resolve format. • TD components meet resolve criteria. • Comms with SCS maintained during resolve.	☐	☐	
q. Report true bearing to SCS, delete the ambiguous bearing lines from the TD (Significant)	☐	☐	
r. After Resolving ambiguity give amplification Report to SCS IAW A5	☐	☐	
s. Ensure Tracker set up for optimal tracking / updating of contact	☐	☐	
2) Conduct Target Motion Analysis:			
a. Determine or asses on the Frequency Observed (Fo) based on: • Intelligence; • Peak High / Peak Low; • CPA; and • Previous holding.	☐	☐	
b. Obtain the following data; • Own ship's course and speed; • Fine Bearing Estimate; and • Frequency received.	☐	☐	
c. Calculate a Line Of Sound diagram to determine the following information: • OAA; and • OSI.	☐	☐	
d. Calculate Frequency Corrected	☐	☐	
e. Determine CTC's relationship to own ship	☐	☐	
f. Correlate CTC relationship using other sources i.e., BB display or acc. Sources	☐	☐	
g. Recommended course/speed alteration if needed	☐	☐	
h. Determine Contacts Bearing Rate and estimate on speed and range based on: • Calculating the difference in degrees of Bearing vs. difference in Time;	☐	☐	

EC 002.02 AMPLIFY TRACK INFORMATION			
• Determining the number of Narrowband / Broadband Beams contact is held on; • Awareness of Oceanographic range predictions; and • Awareness of Oceanographic conditions (i.e. bottom features, composition, etc.).			
3) Supervise Lost Contact Procedures:			
a. When contact is lost, makes Lost Contac Report IAW A5 up to "SCS"	☐	☐	
b. Ensure operator conducts Lost Contact procedure	☐	☐	
c. TAS SUP reports result of Lost CTC procedures	☐	☐	
d. Changes contact status as directed by SCS	☐	☐	

Score: / 31

Trainee must achieve 22 / 31 SATs on non-critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – DIRECT TOWED ARRAY SYSTEM OPERATIONS
PC 002 – DIRECT TOWED ARRAY SYSTEM OPERATIONS

PC 002 – Direct Towed Array System Operations		
Scope	**Location Conditions**	**Instructional Sequence**
PO 002	CANTASS Mission Simulator Trainer or SEACOT.	After EC 002.02
Time	**Classification**	**Instructor**
600 minutes (1 x 50 min period per student)	SECRET	RQ-MS Instructor

Scenario	IAW PC 002 scenario.
Assessment Steps	**Input** – Students conduct this PC in either the CANTASS Mission Simulator or SEACOT, they will be provided with CANTASS console and 1-2 operators, PC 002 written scenario with Line Of Sound Diagram, calculator, reference NAVORD 3137-1, and a subsurface contact the conducts course and/or speed alterations. **Performance** Students will be required to read the scenario, calculate the required towed array speeds/depth/cable scope and provide the instructor with verbal answers. Students will then supervise the CANTASS operator's in: setting up procedures, closing up procedures with voice reports to the SCS/Instructor, direct operators in the search, initial contact procedures, conduct TMA on a target, and supervise lost Contact procedures IAW PC 002 checklist. (600 minutes total, 50 minutes per student) **Output** – Trainee must attain 34 / 48 SATs to pass PC IAW PC 002 checklist. **Feedback** – Student is provided written feedback via the PC 002 checklist and verbally by the instructor upon completion of the PC.

Instructor Resources

1) CANTASS Mission Simulator
2) 2 x CANTASS Operators (1 minimum)
3) PC scenario with Line of Sound diagram template
4) Calculator
5) NAVORD 3137-1 Tow Chart
6) PC 002 Checklist

Note(s) to Instructor

1) PC 002 Practical assessment is designed to evaluate the trainee's ability to Direct Towed Array system operations as the TASSUP IAW PC 002 checklist.
2) Pass mark will be 34 / 48 SATs to pass PC IAW PC 002 checklist.
3) References are permitted.

Assessment Limitations	Nil.
Impact of Failure	Failure to achieve the standard on this PC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this PC, continuation of training without a re-write may be recommended.

PROTECTED B (when complete)

PC 002 DIRECT TOWED ARRAY SYSTEM OPERATIONS			
Item	SAT	UNSAT	Remarks
1) Calculate array towed speeds:			
a. Minimum/Maximum operating depth	☐	☐	
b. Calculate ships speed for array depths (m)	☐	☐	
c. Minimum/Maximum safe depth	☐	☐	
d. Recommend towed array depth	☐	☐	
2) Supervise set up procedures:			
a. Conduct communication check with SCS	☐	☐	
b. Ensure all required equipment is available	☐	☐	
c. Conduct team brief to include: • Exercise/area/situation • Friendly forces • TOI • Environmental conditions • Target detail	☐	☐	
d. Ensure proper sanitization completed, tonal logged	☐	☐	
3) Supervise closing-up procedures:			
a. Set Integration time to include: • Narrow Band (NB) • Broadband (BB) • Demon (DM)	☐	☐	
b. Ensure Verniers are set-up IAW team brief	☐	☐	
c. Ensure array gain is set-up	☐	☐	
d. Adjust Magnetic Variation for AOP	☐	☐	
e. Delegate operator's responsibilities	☐	☐	
f. Ensure tape is recording	☐	☐	
g. Give detailed closing up report to SCS IAW A5	☐	☐	
4) Supervise search, contact procedures:			
t. Ensure detailed NB search is done (SS upper TVM, TBB lower TVM)	☐	☐	
u. Ensure detailed BB search is done	☐	☐	
v. Upon gaining CTC assign operator as "Tracker" and/or "Searcher"	☐	☐	

PC 002 DIRECT TOWED ARRAY SYSTEM OPERATIONS			
w. Ensure searcher sets up to cover full search duties (SS Comb upper TVM and TBB for lower TVM)	☐	☐	
x. Ensure the operator create TGT for TOI	☐	☐	
y. Ensure the operator enters 1 OIT and 1 OUM NB components as directed by the invigilator	☐	☐	
z. Ensure the operator enters TGT source components	☐	☐	
aa. Ensure the operator Searcher enters two BB OUM components on the related tracked TGT, assigning them to the target detail of the target	☐	☐	
bb. Ensure the operator Check for audio on target bearing	☐	☐	
cc. Ensure the operator Generate a CMS330 contact detail for applicable target ensuring all data is in the CCD Contact Detail	☐	☐	
dd. Give detailed Initial Contact report to SCS IAW A5	☐	☐	
ee. Ensure searcher updates applicable CCD data area for CTC i.e., Class. / estimate speed / BB / Audio	☐	☐	
ff. Ensure use of display format; TBB in BRG stabilized on relevant beam and manual input of OUM data	☐	☐	
gg. Ensure logs are accurate i.e., OCS / array data / TGT data and events / environmental conditions / system events	☐	☐	
hh. Conduct detailed amplification report IAW A5	☐	☐	
ii. Ensure the operator resolve for ambiguity: • Correct beams are utilized in proper resolve format. • TD components meet resolve criteria. • Comms with SCS maintained during resolve.	☐	☐	
jj. Report true bearing to SCS, delete the ambiguous bearing lines from the TD (Significant)	☐	☐	

PC 002 DIRECT TOWED ARRAY SYSTEM OPERATIONS			
kk. After Resolving ambiguity give amplification Report to SCS IAW A5	☐	☐	
ll. Ensure Tracker set up for optimal tracking / updating of contact	☐	☐	
5) Conduct Target Motion Analysis:			
i. Determine or asses on the Frequency Observed (Fo) based on: • Intelligence; • Peak High / Peak Low; • CPA; and • Previous holding.	☐	☐	
j. Obtain the following data; • Own ship's course and speed; • Fine Bearing Estimate; and • Frequency received.	☐	☐	
k. Calculate a Line Of Sound diagram to determine the following information: • OAA; and • OSI.	☐	☐	
l. Calculate Frequency Corrected	☐	☐	
m. Determine CTC's relationship to own ship	☐	☐	
n. Correlate CTC relationship using other sources i.e., BB display or acc. Sources	☐	☐	
o. Recommended course/speed alteration if needed	☐	☐	
p. Determine Contacts Bearing Rate and estimate on speed and range based on: • Calculating the difference in degrees of Bearing vs. difference in Time; • Determining the number of Narrowband / Broadband Beams contact is held on; • Awareness of Oceanographic range predictions; and • Awareness of Oceanographic conditions (i.e. bottom features, composition, etc.).	☐	☐	
6) Supervise Lost Contact Procedures:			
e. When contact is lost, makes Lost Contac Report IAW A5 up to "SCS"	☐	☐	
f. Ensure operator conducts Lost Contact procedure	☐	☐	

PC 002 DIRECT TOWED ARRAY SYSTEM OPERATIONS			
g. TAS SUP reports result of Lost CTC procedures	☐	☐	
h. Changes contact status as directed by SCS	☐	☐	

Score: /48

Trainee must achieve 34 / 48 SATs on non-critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 003 – SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT

a. **Performance**: Supervise HMS Operations in a Mine Avoidance Environment (C0005).

b. Conditions:

a. Given:

(1) Mine avoidance environment;
(2) Mine detection team;
(3) Hull Mounted Sonar;
(4) Communication equipment;
(5) Logs and records; and
(6) References.

b. Denied: Nil; and

c. Environment:

(1) At sea;
(2) Day or night; and
(3) Under any weather condition.

c. **Standard**: The Acoustic Junior Supervisor shall supervise HMS Operations in a Mine Avoidance Environment (C0005), to include:

a. Supervising active acoustic analysis (F0025), to include:

(1) Adhering to EMCON (B0025); and

(2) Evaluating correlated acoustic information (F0005).

d. **Training Limitations**: Limited Access Area.

EO 003.01 – SUPERVISE ACTIVE ACOUSTIC ANALYSIS
LG 003.01.01 – SUPERVISE MARINE MAMMAL MITIGATION

LG 003.01.01 – Supervise Marine Mammal Mitigation		
Location Conditions		**Instructional Sequence**
Unclassified classroom		After PC 002
Time	**Classification**	**Instructor**
100 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives		**Instructional Method(s)**
1) Supervise MMM procedures for HMS		1) Student readings 2) 3 star 3) Class discussion
Scenario	Nil.	
Activity Steps	1) Students are provided opportunity to read NAVORD 4003-6 and SEMS SOP E05 (70 minutes – Method 1) 2) Students are to write down the 3 most important pieces of information they gained from the readings. (5 minutes – Method 2) 3) Students are to share their responses with the class and discuss the relevance and implementation of the procedures ship level and in their role as the MS. (25 minutes – Method 3)	
Instructor Resources		**References**
1) 13 x DWAN stations 2) NAVORD 4003-6 3) SEMS SOP E05		1) NAVORD 4003-6 2) SEMS SOP E05 3) NCPM 600
Note(s) to Instructor		
1) Nil.		

EO 003.01 – SUPERVISE ACTIVE ACOUSTIC ANALYSIS
LG 003.01.02 – UNDERSTAND MINE AVOIDANCE PROCEDURES

LG 003.01.02 – Understand Mine Avoidance Procedures		
Location Conditions		**Instructional Sequence**
Unclassified classroom		After LG 003.01.01
Time	**Classification**	**Instructor**
100 minutes	UNCLASSIFIED	RQ-MS Instructor
Learning Objectives		**Instructional Method(s)**
1) Understanding Mines Avoidance SOP (SSOs)		1) Jigsaw 2) 3 Star / Think, Pair, Share
Scenario	Nil.	
Activity Steps	1) Instructor to break students into 4 groups and assign each group ¼ of the Mine Avoidance portion of SSOs. Each group to read and understand, then present their portion to the rest of the class. (45 minutes – Method 1) 2) Students are then to identify the top three items they determined to be important to the MS role. (5 minutes – Method 1) 3) Students are to then pair up and share their responses with their partners. (5 minutes – Method 2) 4) Groups are to then present their responses and class to discuss the impact of SSOs details on the MS role. (45 minutes – Method 2)	
Instructor Resources		**References**
1) DWAN access (x12) 2) Ships Standing Orders Mine Threat Operations (Halifax Class)		1) SSOs
Note(s) to Instructor		
1) Nil.		

EO 003.02 – SUPERVISE ACTIVE ACOUSTIC ANALYSIS
LG 003.01.03 – SUPERVISE HMS OPERATIONS IN MINE AVOIDANCE

LG 003.01.03 – Supervise HMS Operations in Mine Avoidance		
Location Conditions	**Instructional Sequence**	
SEACOT	After LG 003.01.02	
Time	**Classification**	**Instructor**
600 minutes	SECRET	RQ-MS Instructor
Learning Objectives	**Method(s)**	
1) HMS Setup for HMS for mine avoidance 2) Supervise HMS ops in mine avoidance environment	1) EDI 2) Practical (2 x 20 min runs per student)	
Scenario	Nil.	
Activity Steps	1) Instructor to display and explain the following steps to include: a. EMCON policies b. HMS is properly energized and operator system checks completed; c. Communications check on assigned meet-me-net; d. Confirm Logs and equipment is available; and e. Verifying sonar set up status to include: i. HMS to LISTEN; ii. MINE Configuration and accept settings; iii. Ensuring Operators uses proper mine configuration; and iv. Reporting system faults and taking proper initiative to correct. (40 minutes – Method 1) 2) Instructor to display/explain and demonstrate proper operating procedures to include the following: a. Permission to TX from SCS; b. Continuous transmission centered on ships head; c. Recommendations based on Oceanographic conditions; and d. Proper voice reports from HMS operator. (40 minutes – Method 1) 3) Instructor will explain proper voice report procedures identifying COI's to SCS to include the following: a. Initial Report; b. Lost Report; c. Regain Report; d. Resolve Bearing procedures/Report; and e. Course alteration procedure/recommendations. (40 minutes – Method 1) 4) Students are then provided an opportunity to practice minimum of 2 runs, with peers acting as the HMS operator (3-4 contacts per run). Students are provided feedback using the EC/PC 003 checklist. (480 minutes, 2 x 20 minute runs per student – Method 2)		

LG 003.01.03 – Supervise HMS Operations in Mine Avoidance	
Instructor Resources	**References**
1) Mine Investigation / Resolve voice reports 2) HMS Log	1) SSOs 2) SEMS
Note(s) to Instructor	
1) Nil	

EO 003.02 – SUPERVISE ACTIVE ACOUSTIC ANALYSIS
EC 003.01 – SUPERVISE ACTIVE ACOUSTIC ANALYSIS

EC 003.02 – Supervise Active Acoustic Analysis		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.01	SEACOT	After LG 003.01.03
Time	**Classification**	**Instructor**
250 minutes (20 minutes per student)	SECRET	RQ-MS Instructor
Scenario	Nil.	
Assessment Steps	**Input** – Students will conduct this EC in SEACOT, they will be provided with HMS console, 1 HMS operators, the Mine Investigation / Resolve voice reports and HMS Log. **Performance** Students will be required to supervise the HMS Operator in HMS set up in Mine Mode, conducting Marine Mammal Mitigation, Mine Avoidance procedures and active acoustic analysis IAW EC 003.01 checklist. (250 minutes total, 20 minutes per student) **Output** – Trainee must attain 13 / 17 SATs to pass EC IAW EC 003.01 checklist. **Feedback** – Student is provided written feedback via the EC 003.01 checklist and verbally by the instructor upon completion of the EC.	
Instructor Resources		
1) HMS console 2) HMS operator 3) Mine Investigation / Resolve voice reports 4) HMS Log 5) EC 003.01 checklist		
Note(s) to Instructor		
1) EC 003.01 Practical assessment is designed to evaluate the trainee's ability to Supervise Active Acoustic Analysis IAW EC 003.01 checklist. 2) Pass mark will be 13 / 17 SATs to pass EC IAW EC 003.01 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended.	

PROTECTED B (when complete)

EC 003.01 SUPERVISE ACTIVE ACOUSTIC ANALYSIS			
Item	SAT	UNSAT	Remarks
1) Initial HMS Setup – Mine Mode			
a. Conduct Communication Checks (Operator/SCS/bridge)	☐	☐	
b. Conduct surface SITREP	☐	☐	
c. Ensure all required equipment is available to the team and report any deficiencies	☐	☐	
d. Confirm proper setup in "Mine Mode" to include the following IAW A5:			
• HMS in "Listen" (C)	☐	☐	
• Closing up Report to SCS requesting to "TX" (EMCON) (C)	☐	☐	
• Identify all setup faults	☐	☐	
2) Marine Mammal Mitigation			
a. Ensure MMM was conducted IAW SEMS SOP E05 and logged accordingly in the HMS Log	☐	☐	
3) Mine Avoidance Procedures			
a. Requested permission to shine on HMS (if not given during closing up report to SCS)	☐	☐	
b. Ensure transmissions centered on ships heading	☐	☐	
4) Supervise Active Acoustic Analysis			
a. Assist HMS Operator to identify COI's	☐	☐	
b. Reports to include the following IAW SSO's 6.10.6			
• Initial Report	☐	☐	
• Lost Report	☐	☐	
• Regain Report	☐	☐	
• Course Alteration Procedure/Report	☐	☐	
c. Ensure operator records/logs all COIs	☐	☐	
5) Use CODE WORDS IAW APP-7	☐	☐	
Score: /16 **Trainee must achieve 13 / 16 SATs on non-critical factors to pass this EC.**			
Positive Feedback	Improvement Feedback		

EC 003.01 SUPERVISE ACTIVE ACOUSTIC ANALYSIS		
Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 003 – SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT
PC 003 – SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT

PC 003 – Supervise HMS Operations in a Mine Avoidance Environment		
Scope	**Location Conditions**	**Instructional Sequence**
PO 003	SEACOT Trainer	After EC 003.01
Time	**Classification**	**Instructor**
200 minutes (15 minutes per student)	SECRET	RQ-MS Instructor
Scenario	Nil.	
Assessment Steps	**Input** – Students will conduct this PC in SEACOT, they will be provided with HMS console, 1 HMS operators, the Mine Investigation / Resolve voice reports and HMS Log. **Performance** Students will be required to supervise the HMS Operator in HMS set up in Mine Mode, conducting Marine Mammal Mitigation, Mine Avoidance procedures and active acoustic analysis IAW PC 003 checklist. (200 minutes total, 15 minutes per student) **Output** – Trainee must attain 13 / 17 SATs to pass PC IAW PC 003 checklist. **Feedback** – Student is provided written feedback via the PC 003 checklist and verbally by the instructor upon completion of the PC.	
Instructor Resources		
1) HMS console 2) HMS operator 3) Mine Investigation / Resolve voice reports 4) HMS Log 5) PC 003 checklist		
Note(s) to Instructor		
1) PC 003 Practical assessment is designed to evaluate the trainee's ability to Supervise HMS Operations in a Mine Avoidance Environment IAW PC 003 checklist. 2) Pass mark will be 13 / 17 SATs to pass PC IAW PC 003 checklist. 3) References are permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this PC may result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this PC, continuation of training without a re-write may be recommended.	

PROTECTED B (when complete)

PC 003 SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT			
Item	SAT	UNSAT	Remarks
1) Initial HMS Setup – Mine Mode			
a. Conduct Communication Checks (Operator/SCS/bridge)	☐	☐	
b. Conduct surface SITREP	☐	☐	
c. Ensure all required equipment is available to the team and report any deficiencies	☐	☐	
d. Confirm proper setup in "Mine Mode" to include the following IAW A5:			
• HMS in "Listen" (C)	☐	☐	
• Closing up Report to SCS requesting to "TX" (EMCON) (C)	☐	☐	
• Identify all setup faults	☐	☐	
2) Marine Mammal Mitigation			
b. Ensure MMM was conducted IAW SEMS SOP E05 and logged accordingly in the HMS Log	☐	☐	
3) Mine Avoidance Procedures			
c. Requested permission to shine on HMS (if not given during closing up report to SCS)	☐	☐	
d. Ensure transmissions centered on ships heading	☐	☐	
4) Supervise Active Acoustic Analysis			
d. Assist HMS Operator to identify COI's	☐	☐	
e. Reports to include the following IAW SSO's 6.10.6			
• Initial Report	☐	☐	
• Lost Report	☐	☐	
• Regain Report	☐	☐	
• Course Alteration Procedure/Report	☐	☐	
f. Ensure operator records/logs all COIs	☐	☐	
5) Use CODE WORDS IAW APP-7	☐	☐	
Score: /16 **Trainee must achieve 13 / 16 SATs on non-critical factors to pass this PC.**			
Positive Feedback	Improvement Feedback		

PC 003 SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT		
Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

a. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

b. All ECs are formative assessments that help the trainee and the instructors to recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

c. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

d. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

e. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

f. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

g. Continuous monitoring of trainee progress is required to provide the following:

h. Early warning of trainee difficulties;

i. Feedback on the effectiveness of training;

j. Information for the Training Review Board (TRB) (if required); and

k. Information to assist in the completion of the CF 377 Course Report.

l. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

m. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

n. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

o. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

p. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

q. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

r. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defense Subject Classification and Disposition System (DSCDS) Primary Number 4500.

ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A – CAF or other Official Canadian Government Publications		
A1	CFCD 113 Vol 1	Maritime Command Jezebel Analysis Manual
A2	CFCF 113 Vol 2	Maritime Command Acoustic Information Guide (AIG)
A3	C69-746-000/MS-001	Towed Array Group SQR-19
A4	CFCD 106	Maritime Tactical Instructions
A5	NCPM 600	UWW Procedural Manual for HALIFAX Class
A6	NAVORD 3137-1	Canadian Towed Array Sonar and NIXIE Towing and Operating Instructions
A7		Ship Standing Orders (SSO AL10.1) Chapter 6.10.6.1-6.8 Mine Operations
A8	NOSTP 201	REL VEL Handbook
A9	NAVORD 3470-1	Maritime Acoustic Security Policy
A10	C-69-748-000/MB-001	Operators Manual CANTASS Book 2
A11	C-69-799-000/MB-001	Operating Instructions Detecting-Ranging Set, Sonar, AN/SQS-510
A12	C-69-746-000/MS-001	Towed Array Group
A14		MILOC, Military Oceanographic Manual
A15	NAVORD 2000-2	Request for Intelligence Process
A16	C-10-688-000/MB-001	SEACOT Operating Instructions
A17		Halifax Class Safety and Environmental Management System (SEMS) Manual
A18	NAVORD 4003-6	Marine Mammal Mitigation Procedures for Active Sonar Use
B – Official Allied Force and Allied Government Publications		
B1	ATP 28	(Classified)
B2	ATP 1	Allied Tactical Publication
B3	RP33	Fleet Oceanographic and Acoustic Reference Manual
B4	APP-7	
C – Commercial Publications		
C1		CANTASS Users' Manual For Acoustic Operators
D – Other Publications		

ANNEX B – TASK LIST

This table provides an overview of MTL tasks WRT if/where they are trained within this QSP. No train substantiations are located in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
A0005	Perform communications circuit functional checks (comm checks)	NT	
A0010	Operate communications equipment	NT	
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	NT	
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	NT	
B0005	Perform pre-sail checks on equipment	NT	
B0015	Maintain state boards	NT	
B0025	Adhere to Emission control (EMCON) restrictions	T	PO 003
B0030	Contribute to intelligence gathering Ops	T	PO 002
B0035	Display intelligence via UWW brief	NT	
B0040	Disseminate intelligence information	NT	
B0045	Identify Capacities and limitations of Platforms	T	PO 001
B0050	Operate CS	NT	
B0055	Maintain operational logs	NT	
C0005	Supervise HMS Operations in a Mine Avoidance Environment	T	PO 003
C0010	Supervise Mine Detection Team in a Mine Avoidance Environment	NT	
C0020	Operate passive acoustic equipment	NT	
C0025	Operate ancillary acoustic equipment	NT	
C0030	Operate oceanographic equipment	NT	
C0040	Analyzing XBT/XSV Forms	NT	
C0060	Monitor sonobuoys	NT	
C0065	Analyze passive acoustic data	NT	
C0070	Track acoustic contacts	T	PO 002
C0075	Annotate recording media	NT	PO 002
C0080	Classify passive acoustic data	NT	
C0085	Localize passive acoustic contacts	T	
C0095	Analyze TMA data	T	PO 002
C0105	Correlate active/passive/non-acoustic information	NT	
C0115	Calculate acoustic sensor range information	NT	
C0120	Calculate array tow speeds, tilt and/or balancing	T	PO 002
C0125	Analyze acoustic tapes/multi-media	NT	
C0127	Ensure Tow Array Sonar System is set up	T	PO 002
C0135	Perform noise rounds	NT	

C0140	Coordinate the employment of UWW weapons and sensors	NT	
C0150	Determine Oceanographic Effects on Acoustic Operations	NT	
C0155	Direct Towed Array System Operations	T	PO 002
C0160	Establish capabilities and limitations of towed array operations in operation areas (OPAREAS)	T	PO 002
C0170	Assist with loading/handling of torpedoes	NT	
C0185	Charge torpedo tubes/flask	NT	
C0195	Perform Passive Aural Listening (PAL)	T	
D0025	Perform UWW weapon firings	NT	
D0030	Perform weapons system misfire procedures	NT	
D0050	Assess UWW threat information	T	PO 003
D0080	Interpret LOFAR Gram Acoustic Characteristics	T	
E0005	Compile ASW tactical picture	T	PO 001
E0020	Amplify track information	T	PO 002
F0005	Evaluate correlated acoustic information	T	PO 002
F0010	Record acoustic intelligence data	T	PO 002
F0015	Identify changes in contact's readiness and tactical states	T	PO 002
F0025	Conduct active acoustic analysis	T	PO 003
F0030	Conduct mission reconstruction	NT	
F0035	Perform basic passive acoustic analysis	NT	
F0045	Interpret oceanographic data	T	PO 002
G0010	Plan / design training exercises	T	PO 001
G0045	Assist in individual training events	T	PO 001
G0055	Correct individual training deficiencies	NT	
G0060	Assist in team training events	T	PO 001
G0065	Conduct team training events	T	PO 001
G0075	Conduct classroom instruction/training	NT	
G0080	Conduct continuation / proficiency training	NT	
G0090	Input training scenarios	T	PO 001
G0095	Operate simulator/trainer systems	T	PO 001
Added Tasks			

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Dates: 02 May 2013

Location: CFNOS Halifax

QSP Version 1.3 HCM Hybrid

a. These changes are designed to make this QSP HCM Hybrid compliant and ensure capabilities and limitations of new HCM driven equipment are covered. The changes were fully discussed and drafted during the SONAROP HCM QSP working group held 29 Apr – 30 May. In attendance were:

    a. PO1 Couture (CFNOS Procedures instructor)
    b. PO1 Jenkins (Senior Sonar OP Advisor Future Systems Training
    c. Development (FSTD)
    d. PO1 MacDowall (CFNOS Senior SONAROP Quality Assurance Rep.)
    e. PO1 Stewart (CFNOS SONAROP Senior instructor)
    f. PO2 Fisher (CFNOS SONAROP Quality Assurance Rep.)

b. The Only change included in HCM Hybrid Version 1.0 was the inclusion of "An Introduction to Electronic TMA in CMS330". This inclusion is found in EO 004.1

Dates: 11 July 2013

Location: CFNOS Halifax

QSP Version 1.4 HCM Compliant

c. These changes are designed to make this QSP fully HCM compliant. The changes were fully discussed and drafted during the SONAROP HCM QSP working group held 29 Apr – 30 May and approved 11 July 2013. In attendance were:

    a. PO1 Couture (CFNOS Procedures instructor)
    b. PO1 Jenkins (Senior Sonar OP Advisor Future Systems Training
    c. Development (FSTD)
    d. PO1 MacDowall (CFNOS Senior SONAROP Quality Assurance Rep.)
    e. PO1 Stewart (CFNOS SONAROP Senior instructor)
    f. PO2 Fisher (CFNOS SONAROP Quality Assurance Rep.)

d. Changes from HCM Hybrid QSP to the HCM Steady State QSP were all administrative in nature. Specifically:
   a. Change all references to Shipboard Standard Display to read "Multi-Function Workstation (MFW)".
   b. Change all references to COT and COT to read "Combat Operations Trainer".
   c. Change all references to OSF to read "Operator Selectable Functions"

QSP Version 1.5 Minor Amendments

Dates: 20 August 2015

e. Secretarial changes were made to reflect the new JBS Course Title and National Qualification Code. Minor spelling amendments and old references to CCS that had been overlooked in previous HCM QSP amends, were also corrected.

QSP Version 2.0 Modularize course

Dates: 27 Feb – 31 Mar 2017

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Devine	S.J.	Lt(N)	00207	TDC(A)
Matte	K.M.L.	SLt	00211	TDC(A)
Fisher	W.J.	PO2	00324	TDC(A)
Plourde	A.P.	MS	00324	TDC(A)
Rushton	J.O.	MS	00324	TDC(A)

Working Group Objective:

f. The purpose of convening this working group was to:
   a. Shorten the course;
   b. Ensure all aspect of the course is performance-based and align with the current war;
   c. Modularize; and
   d. Accomplish all design and modifications required in the 30 days allocated time.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform comm Circuit functional checks	This task is trained on RQ-S1
A0010	Operate comm equipment	This task is trained on RQ-S1
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	This task is trained on RQ-PO2
A0035	Extract info from COMM Plan, EMCON Plan and FREQ Plans	This task is trained on RQ-PO2
B0005	Perform pre-sail checks on equipment	This task is no longer performed by SONAR Operators onboard ship.
B0015	Maintain stateboards	This task is trained on RQ-S1
B0035	Display intelligence via UWW brief	This task is trained on RQ-S3 & S1
B0040	Disseminate intelligence information	This task is trained on RQ-S1 & PO2
B0050	Operate CS	This task is trained on RQ-S1, there is no specific CS role for RQ-MS
B0055	Maintain operational logs	This task is trained on RQ-S3
C0010	Supervise Mine detection Team in a mine avoidance environment	This task is trained on RQ-PO1
C0020	Operate passive acoustic equipment	This task is trained on RQ-S3
C0025	Operate ancillary acoustic equipment	This task is trained on RQ-S3 & S1
C0030	Operate oceanographic equipment	This task is trained on RQ-S3 & S1
C0040	Analyzing XBT/XSV Forms	This task is trained on RQ-S3 & S1
C0060	Monitor sonobuoys	This task is trained on RQ-S1
C0105	Correlate active/passive/non-acoustic information	This task is trained on RQ-PO2
C0115	Calculate acoustic sensor range information	This task is trained on RQ-S1
C0125	Analyze acoustic tapes/multi-media	This task is job specific to ADAC.
C0135	Perform noise rounds	This task is trained on RQ-S3
C0150	Determine Oceanographic Effects on Acoustic Operations	This task is trained on RQ-S1 & PO2.
C0170	Assist with loading/handling of torpedoes	This task is trained on RQ-S3
C0185	Charge torpedo tubes/flask	This task is trained on RQ-S3
D0025	Perform UWW weapon firings	This task is trained on RQ-S3
D0030	Perform weapons system misfire procedures	This task is trained on RQ-S3
F0035	Perform basic passive acoustic analysis	This task is trained on RQ-S3
F0045	Interpret oceanographic data	This task is trained on RQ-S1/PO1

Discussions / Decisions:

g. PO 001 Analyze Passive Acoustic Data is taught by ADAC and is covered under it owns QSP: Royal Canadian Navy Advanced Acoustic Analysis Course (AAAC) QSP – 118722. After discussion, the members have decided to keep this PO embedded in the RQ-MS course for fleet support reasons. By keeping the AAAC in the RQ-MS, the candidates have to leave ship only once. Also, candidates without the AAAC qualification can still be loaded on the RQ-MS. Therefore, the QSP format regarding PO 001 and the related EOs had to be significantly modified in order to fit this vision.

h. PO 003 Supervise the Localization of Passive Acoustic Contacts by Target Motion Analysis (TMA) that involves plotting the different types of plots and some calculations. TMA plotting should be removed from this course as they are completed through CMS 330, however TMA calculations, relationships and Doppler can still be provided as amplifying information.

i. PO 002 was developed out of a need to train RQ-MS SONAR OP to perform the duties of Game Piece Operators onboard HMC Ships and Training Establishments. This PO was also used to achieve the Plan/Design Training events task required for the RQ-MS level.

j. PO 003 HMS Mine operation tasks were separated from previous version as they have no similarities to performance of TASSUP and are now allocated in PO 004.

k. Modularization of content will enable easier recourse making training more efficient in the same process by enabling Training Establishments (TEs) to load only the trainees that haven't completed the AAAC qualification to this module. Therefore, the candidates that have this qualification will only attend the Module 2 portion of this course.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Develop MLPs as per Chapter 4	PO2 Fisher	31 Mar 2017
2	Develop TEL as per Spiral Annex	TBD	ASAP

Course History:

l. Several lessons were learned and recommendation made from the pilot course, which was conducted from 11 Feb until 1 Apr 05. In July 05 amendments were finalize. The following amendments were initiated as "S-1" to improved Chapter 1 & 2:

Course length Calculation changes were recommended as follow: Actual Training Days + Trainer Days (* see Note in "Course Capacity")	21
Admin Day (if required)	1
TAT	2
Actual training Days at ADAC (including TAT)	15
Course Duration (Total) (* see Note in "Course Capacity")	39

(1) **Justification**: The table in the original document was as follows:

(a) "Actual Training Days + Training Days = 32
Admin Day (if required) = 1
TAT = 2
Course Duration (Total) = 35"

(2) The Portion of the course taught at ADAC, the Advanced Acoustic Analysis Course (AAAC), requires to be accounted for individually and is a full 15 days. The AAAC meets all requirements for the modified PO 002.

(3) The Course Duration has been lengthened based on pilot course evaluation with additional training time required for TMA and CANTASS phases. The training time is based on optimum student load. If the student load exceeds the optimum number additional training days will be required.

b. Course Capacity was modified from minimum 6, optimum 9 and maximum 12 to minimum 4, optimum 6 and maximum 12

(1) **Justification**: All numbers were changed because of loading restrictions and limitations. When students from the SONAR OP QL5B course are loaded, then liaison and coordination must be made between DMTE, ADAC, CFNOS and CFFSE in order to coordinate the serials. The usual process to load the AAAC is to guarantee 6 positions for the QL6A and they are loaded by DMTE. ADAC reserves the other 6 student positions for other units as dictated by their requirements. However, if the other 6 positions are not filled DMTE could load additional students on short notice. Upon liaison between CFNOS, CFFSE, ADAC and DMTE, it was agreed that if more than the optimum number of students are to be loaded on a course, all training establishments involved and DMTE need to have consultation and be in agreement.

c. Information about ranking and grading was removed from chapter 1:

(1) **Justification:** A new DAOD was expected to be promulgated in 2005 that should provide directives on grading and ranking. CFNOS will adhere with directives and information will be changed to reflect them in Chapter 3. It will allow CFNOS to modify grading and ranking to allow consistency with other courses and school policy as published in CFNOS Standing Orders. Also, because this course will be taught at CFFSE (with a phase being taught at ADAC (P)) and CFNOS (with a phase being taught at ADAC (A)), chapter 3 will be expanded to cover all important aspects on grading, ranking, PRB, TRB, student progression, EC and PC administration as different training establishments may have different Standing Orders. The recommendation to instruct the course on both coast was initiated from the SONAR OP ADVISORY GROUP MEETING held in Halifax in 2004.

d. Qualification "AEJR" from the AAAC was remove from chapter 1:

(1) **Justification**: Based on SONAR OP advisory group meeting held in Esquimalt 5-6 May 05 it was agreed that only ALBC will be granted. Once a SONAR OP has the ALBC competency, it must be understood by all units that this includes AEJR competency. QL5Bs should only have the ALBC competency and AEJR should not include the RQ-SONAROP-MS SONAR OP as it is now CORE Training. The SONAR OP JBOS now reflects tasks and knowledge from AEJR (AAAC).

e. Re-alignment of all POs in Chapter 2 were recommended as follow:

(1) PO 401 to be change for PO 001; PO 402 to be removed from QSP, as it will be incorporated in PO 403; PO 403 to be change for PO 002; PO 404 to be change for PO 003; and PO 405 to be change for PO 004.

(2) **Justification**: - PO 402 and PO 403, which were all information taught on AAAC, were combined and renamed to PO 002. Combining all information in PO 002 helped re-align all information in Chapter 4, which was re-organized to better match the sequencing and teaching points as they are taught on the AAAC. The QSP was also modified to adhere to QSP formatting. It included:

(a) renaming all PO # to 000 series instead of 400 series;
(b) moving all annexes after chapter 4 and renaming all annexes according to QSP template; and
(c) correct and fill up all information in annexes.

m. The pilot course had several scheduling issues. Lessons learned were as follow:

a. There is only one way that this course can be scheduled. The training establishments and DMTE have to be in close liaison and ensure that the following occurs:

(1) Students have to start the course 3 training days prior to start the AAAC phase conducted at ADAC. It allows for ½ day administration followed by the first PO "001" which is 2 and ½ day.

(2) Students conduct the AAAC phase, which starts on Monday and last 15 days. ADAC requires a full 15 days to conduct instruction and this phase as to be accounted as such instead of periods;

(3) The subsequent phases have to be in order, which is PO 003 then PO 004. It allows for the best flow of instruction.

n. Several changes were also made to Chapter 4 to reflect all changes made to Chapter 2. Adjustment and re-alignment was made to all periods to reflect lessons learned from the pilot course. Recommendations for steady state course were made based on conditional approval of all previous recommendations.

o. 23 May 2012 – Administrative amendments to reflect the Command name change and the establishment of new lines of authority.

p. Additional amendments were recommended in spring 2006, after the second serial held in Victoria B.C, Oct-Dec 2005.

a. Course length Calculation (IAW MARCORD 9-30) to be replaced by:

Actual Training Days + Trainer Days (* see Note in "Course Capacity")	21.6
Admin Day (if required)	2
TAT (10 periods)	1.4
Actual training Days at ADAC (including TAT)	15
Course Duration (Total) (* see Note in "Course Capacity")	40

(1) ***Justification***: The table in the original document was as follows:

(a) *Actual Training Days + Training Days* = *21*
Admin Day (if required) = 1
TAT = 2
Course Duration (Total) = 39"

(2) The Course Duration was lengthened because additional training time was required for the HMS Supervisor phase, and the addition of 1 Admin day. The additional Admin day was added to allow for the PO conducted at ADAC in conjunction with the AAAC to be first. By moving PO 001 "Analyze and Classify Passive Acoustic Sources" at the beginning of the course, it will allow for any trainee who successfully challenge this PC for this PO to join the course 3 weeks later instead of having to report to the course for 3 days then having to wait out the 3 weeks while that PO is being instructed. The training time is based on optimum student load. If the student load exceeds the optimum number additional training days will be required.

b. Introduction of PO 001 (Passive Acoustics Phase) Challenge test:

(1) **Justification**: The MOC advisory group requested for a waiving rule to be added for exemption of people who previously had the AEJR. After investigation of many alternatives and considerations, CFNOS Stds staff came to the conclusion that a "Challenging Test" would be the best and only way to proceed. With this test there will be no doubt that the trainee can be exempt if he or she pass the challenging PC. ADAC Halifax agreed to generate the challenging PC and CFNOS would ensure that the test complies with QSP. This test will also ensure the trainee current with the latest acoustic threat and analysis skills prior of being granted PO 001. If the trainee passes the PC, then trainee will be able to report for the RQ-SONAROP-MS course 3 weeks later and bypass PO 001 (which is the portion being taught at ADAC in conjunction with AAAC). It would also allow enough time for ADAC to find a suitable replacement on course.

c. Re-alignment of PO 001 and PO 002 in Chapter 2

(1) **Justification**: with the introduction of the challenge test, a trainee that successfully completes the test will be credited with the training portion of the course conducted at ADAC. By placing the portion instructed at ADAC up front, it will enable the trainee to remain at his/her unit until joining the course at the beginning of the next Performance Objective.

d. The introduction of the Best Overall Student Award instead of the Top Student Award. The Top Student Award used to lead to confusion since there is no standing or ranking allocated for this course. The Best Overall Student Award is a compilation of the trainee's marks and votes by his/her peers. The award was originally added to the course to promote "Esprit de Corps" and teamwork as well as encourage trainees to do their best.

e. 2 new periods were added to first HMS EO to cover new Teaching Points (Marine Mammal Mitigation ramp-up procedures and SCS console familiarization) and 1 period was added to the HMS PC to allow enough time for de-briefing. This last period was taken from the development phase of the third HMS EO that 9 periods were originally allocated to it, however it was observed that 8 periods was sufficient. When all periods were accounted for, it changed the length of HMS PO from 17 to 19 periods.

f. "Industrial Sources noise" was added as a new Teaching Point for rename EO 001.1. No extra time was required for this EO to cover the new subject.

g. All PC checklists were amended to enhance the efficiency of the tests and also to put all PC in a standardized format by including the assessment sheet at the end of each PC.

h. The Suggested Sequencing and Timing Table was amended because of the switch between PO 001 and PO 002, and to reflect the addition of periods for the HMS PO.

i. Paragraph 20 in Chapter 3 was modified to add clarity and ensure that since the PC conducted at ADAC has two parts, if on the first attempt the student does not pass the analysis portion but pass the classification portion then on the rewrite the opposite happens and the student pass the analysis but fails the classification, the results from the different test cannot be combined together taking the best portions of each test.

j. After "Div Comd" in para 21, 28 and 29 from Chapter 3, "or his/her delegated representative" was added to allow Training Establishments to have the Div Comd or delegated representatives to be the authority for PRBs and/or TRBs. This will provide more flexibility to TEs as long as it is clearly identified who the authority was delegated to.

k. In paragraph 29 from Chapter 3, the term "MOC" was amended with "MOS" as the term MOC is no longer valid.

l. Paragraph 33 from Chapter 3 was modified to ensure that if a TRB is convened concerning the PO conducted at ADAC then the ADAC Course Officer or assigned representative will be part of the TRB.

m. Paragraph 34, 35 and 36 from Chapter 3 was modified in order to provide further directives on the CF 377 course report. After further analysis of the course and the PC, it was decided that for the TASSSUP course there would be NO ranking, NO grading and NO class standing. The only PC that might accommodate a good evaluation with grading is the TMA PC. It was determined that grading was relative with all other PC. Therefore, this course will be only pass or fail. However, the narrative block of the CF 377 will amplify the strengths and weaknesses of the students. The BEST OVERALL STUDENT AWARD will also be narrated for the student who receives it. The narrative will exclude any kind of grade. Also, the tables following PC CHECKLISTS with "Unsatisfactory, below standards, satisfactory, above standard and exceed standard" were removed from the QSP as they were misleading. For this course, students have to meet the standard to pass the course.

n. Some other changes were done to checklist and the actual names of references were removed and replace with references instead.

o. The type of test for EC 004.2 was modified and "multiple choice" was removed to better reflect the type of test used for the EC.

p. "TASSUP" acronyms were replaced with "TASSSUP" through the document as Towed Array Sonar System Supervisor is abbreviated with TASSSUP not TASSUP.

q. ADAC is CCA for the AAAC which is incorporated into the Sonar Op QL5B Course. As ADAC has amended their QSP and our QSP PO 001 is an exact copy, an amendment was required. There is no longer a requirement for a PC 001 theory exam, therefore the QL5B QSP required an amendment (dated 06 Jun 11) to reflect the AAAC QSP.

<u>QSP Version 2.1 Post Combat Training Review</u>

<u>Dates</u>: 30 Apr 2019

<u>Location:</u> Naval Training Development Centre (Atlantic)

r. PO 001 Analyze Passive Acoustic Data is taught by ADAC and is covered under it owns QSP: Royal Canadian Navy Advanced Acoustic Analysis Course (AAAC) QSP – 118722. During the CTR, ADAC Training PO determined that additional training time was required for the AAAC portion to accommodate new instructional methodology and updated course content. This also meets the RCAF requirements for the AAAC. As a result, an additional 2 training days were allocated for the course and the QSP content was updated to meet ADAC requirements.

s. PO 002 was developed out of a need to train RQ-MS SONAR OP to perform the duties of Game Piece Operators onboard HMC Ships and Training Establishments. This PO was also used to achieve the Plan/Design Training events task required for the RQ-MS level.

t. PO 003 Supervise the Localization of Passive Acoustic Contacts by Target Motion Analysis (TMA) that involves plotting the different types of plots and some calculations. TMA plotting should be removed from this course as they are completed through CMS 330, however TMA calculations, relationships and Doppler can still be provided as amplifying information.

u. PO 003 HMS Mine operations tasks were separated from previous version as they have no similarities to performance of TASSUP and are now allocated in PO 004.

QSP Version 2.2 RCN JR Rank Change

Dates: 17 November 2020

Location: Naval Training Development Centre (Atlantic)

v. IAW CANFORGEN 112/20 RCN JR Rank Designation Change 041200Z SEP 20, all previous RCN JR Rank designations in this QSP have been amended to reflect the new policy.

w. Amendments made to the lesson Enabling Objective naming convention. Changed from Number with Letter to just number. This was to align with NFS Scheduling program requirements.

x. Addition of previously omitted Chapter 3: Assessment of Trainee.

QSP Version 3.0 Conversion of Legacy to 3 Chapter format

Dates: 23 March 2022

Location: Naval Training Development Centre (Atlantic)

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Meyers	R	Lt	00211	NTDC(A)
Tallman	J.R.	CPO2	00324	NTDC(A)
Blonde	ADJ	MS	00324	NTDC(A)
Deneau	JA	PO2	00324	NTDC(A)
Lavallee	TJ	MS	00324	NTDC(A)
Craig	B	MS	00324	NFS(A)

y. The SONAR Operator RQ-MS QSP has been in the Legacy format with separate Master Lesson Guidance. With NTDC moving towards the new 3 Chapter QSP format with the Master Lesson Guidance incorporated into the QSP, there was a need to convert the Legacy QSP into the new format and add the Master Lesson Guidance.

z. There was also an issue that the RQ-MS QSP was modularized into 2 separate modules, with Mod 1 being the Advanced Acoustic Analysis Course and Mod 2 being the RQ-MS QSP focusing on ASW equipment and procedures. In the last iteration of the QSP, Mod 1 (AAAC) was incorporated into the RQ-MS QSP as PO 001 Analyze Acoustic Data, however this caused a duplication of training as the NPTG HQ Training Schedule often had the AAAC scheduled before the RQ-MS course, resulting in the same Performance Objective, Analyze Acoustic Data, being instructed twice, back to back. The QSP Working Groups goal was to remove PO 001 Analyze Acoustic Data from the RQ-MS QSP and indicate in Chapter 1 that the AAAC was to be a pre-requisite for all RQ-MS trainees. This will rectify the duplication issue, but will require the QSP to conduct a Major Amendment IAW NPTORD 4500-22.

aa. Working Group Objective:

The mandate of the QSPWB was to:

a. Review the Legacy SONAR OP RQ-MS QSP and Master Lesson Guidance,
b. Removal of PO 001 Analyze Acoustic Data
c. Convert the Legacy QSP to the new 3 Chapter QSP format via a contracted technical writer;
d. Amend Chapter 1 to indicate AAAC (AEJR) as a pre-requisite for the course;
e. Review all Master Lesson Guidance;
f. Update all subsequent Annex's; and
g. Submit the converted QSP to NPTG HQ for approval.

33. <u>Specification Task – No Train Decisions:</u>

Annex B contains all tasks associated with SONAR OP RQ-MQ OSS. All tasks shall be trained.

Task #	Task Statement	Substantiation
A0005	Perform comm Circuit functional checks	This task is trained on RQ-S1
A0010	Operate comm equipment	This task is trained on RQ-S1
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	This task is trained on RQ-PO2
A0035	Extract info from COMM Plan, EMCON Plan and FREQ Plans	This task is trained on RQ-PO2
B0005	Perform pre-sail checks on equipment	This task is no longer performed by SONAR Operators onboard ship.
B0015	Maintain stateboards	This task is trained on RQ-S1
B0035	Display intelligence via UWW brief	This task is trained on RQ-S3 & S1

B0040	Disseminate intelligence information	This task is trained on RQ-S1 & PO2
B0050	Operate CS	This task is trained on RQ-S1, there is no specific CS role for RQ-MS
B0055	Maintain operational logs	This task is trained on RQ-S3
C0010	Supervise Mine detection Team in a mine avoidance environment	This task is trained on RQ-PO1
C0020	Operate passive acoustic equipment	This task is trained on RQ-S3
C0025	Operate ancillary acoustic equipment	This task is trained on RQ-S3 & S1
C0030	Operate oceanographic equipment	This task is trained on RQ-S3 & S1
C0040	Analyzing XBT/XSV Forms	This task is trained on RQ-S3 & S1
C0060	Monitor sonobuoys	This task is trained on RQ-S1
C0065	Analyze passive acoustic data	This task is trained on AAAC
C0080	Classify passive acoustic data	This task is trained on AAAC
C0085	Localize passive acoustic contacts	This task is trained on AAAC
C0105	Correlate active/passive/non-acoustic information	This task is trained on RQ-PO2
C0115	Calculate acoustic sensor range information	This task is trained on RQ-S1
C0125	Analyze acoustic tapes/multi-media	This task is job specific to ADAC.
C0135	Perform noise rounds	This task is trained on RQ-S3
C0150	Determine Oceanographic Effects on Acoustic Operations	This task is trained on RQ-S1 & PO2.
C0170	Assist with loading/handling of torpedoes	This task is trained on RQ-S3
C0185	Charge torpedo tubes/flask	This task is trained on RQ-S3
C0195	Perform Passive Aural Listening (PAL)	This task is trained on AAAC
D0025	Perform UWW weapon firings	This task is trained on RQ-S3
D0030	Perform weapons system misfire procedures	This task is trained on RQ-S3
D0080	Interpret LOFAR Gram Acoustic Characteristics	This task is trained on AAAC
F0035	Perform basic passive acoustic analysis	This task is trained on RQ-S3
F0045	Interpret oceanographic data	This task is trained on RQ-S1/PO1
Task #	Task Statement	Substantiation
A0005	Perform comm Circuit functional checks	This task is trained on RQ-S1
A0010	Operate comm equipment	This task is trained on RQ-S1

Minutes:

10 – 28 January 2022

- Informed NPTG HQ Training Manager (CPO2 Stewart) of intent to convert AAAC QSP to new format

- RQ-MS QSP reviewed by NTDC(A) CCD Analysts to determine requirement for any major and/or minor amendments
- A Major Amendments was identified, requirement to remove PO 001 Analyze Acoustic Data due to duplication of exact same Performance Object found in the Advanced Acoustic Analyst Course (AAAC).
- Chapter 1 Prerequisites was amended to indicate the requirement to complete the AAAC NQUAL AEJR prior to achieving the SONAR OP RQ-MS NQUAL ALBC
- RQ-MS Master Lesson Guidance reviewed and amended by MS Blonde
- Drafted DsAR for submission by CoC to conduct Major Amendment of RQ-MS QSP

09 February 2022
- Submitted RQ-MS QSP and Master Lesson Guidance to NTDC(A) Civilian contractor Susannah Daines for conversion to 3 Chapter format

25 February 2022
- Received RQ-MS QSP back from technical writer converted to the new QSP format and direction on what was still required to be updated / completed

09 March 2022
- Conducted a RQ-MS QSP Working Group with NFS(A) staff to review recommended changes by TDO and Technical Writer

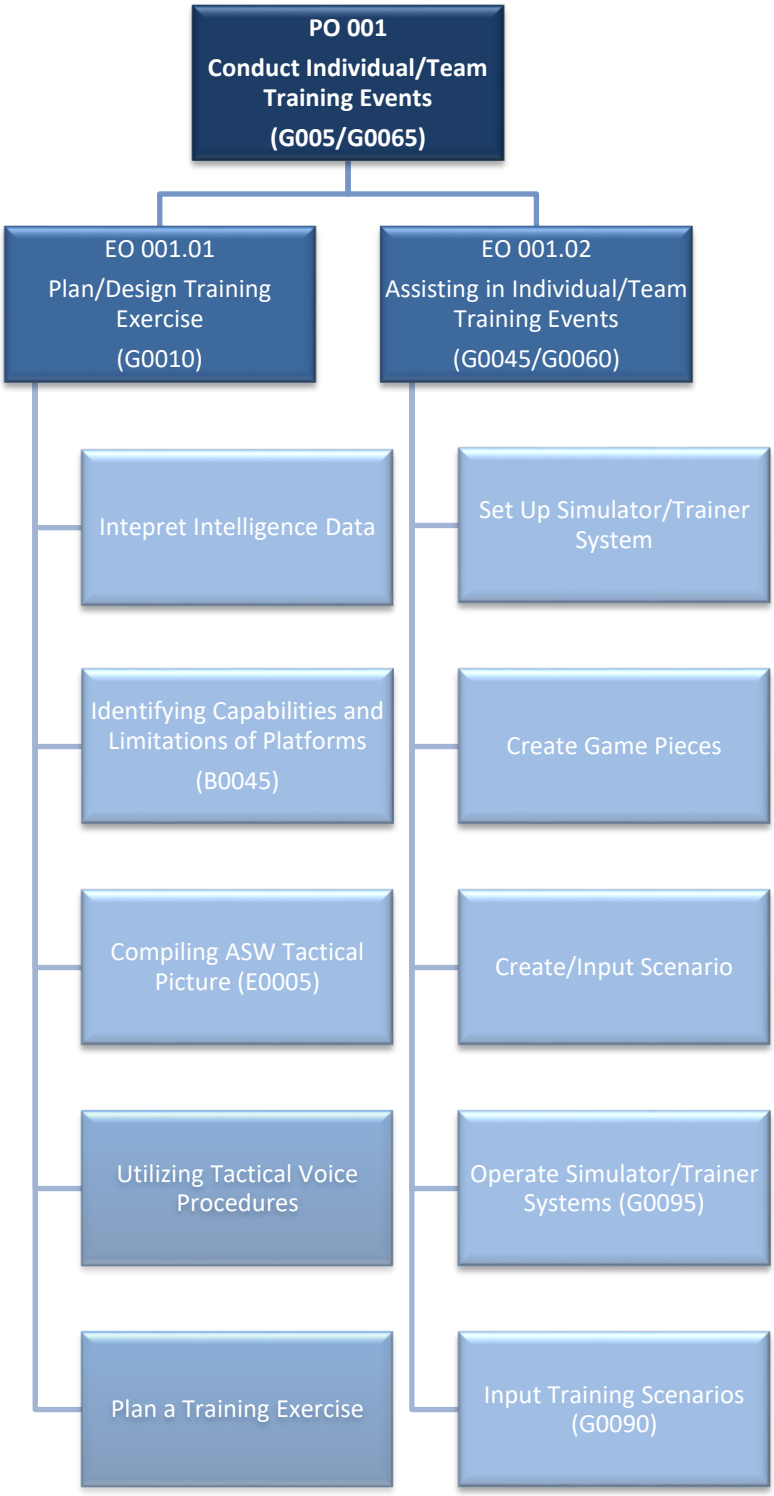
12 April 2022
- Additional review of the amendments to the Master Lesson Guidance were conducted with support from NFS(A) Instructor MS Brian Craig
- It was identified by the NFS(A) instructor that during a number of the ECs and PCs that although there was sufficient time based on a minimum course load, which is often the case with this course, there would be insufficient time with an optimum or maximum course load that that additional time should be added to allow for future courses with optimum and maximum course loads.
- It was identified in PO 001 that during ECs and the PC, that due to only having 1 instructor and 1 console available, it placed difficult time constraints on the instructor resulting in rushed EC/PC's with limited or no time allotted for trainee debrief. An additional 500 minutes or 10 periods were added to the ECs and PC to allow for sufficient time based on maximum course load.
- It was identified in PO 002 that there was no EC 002.02 prior to PC 002, this had been previously removed based on redundancy, however it was determined that EO 002.02 should have an EC, it was created and an additional 500 minutes was allotted for EC 002.02
- Additional time was also allocated for EC 002.01 of 150 minutes to allow for maximum course load
- Additional time was allocated for Lesson Guidance within this section due to instructor feedback that there was not sufficient practice time due to the aging CANTASS Mission Simulator having IT issues that often reduced practical training time. Overall all PO 002 was allotted an additional 1000 minutes or 20 periods of ATT
- It was identified in PO 003 that there was no EC 003.01 prior to PC 003, this had been previously removed based on redundancy, however it was determined that EO 003.01 should have an EC, it was created and an additional 250 minutes was allotted for EC 002.02

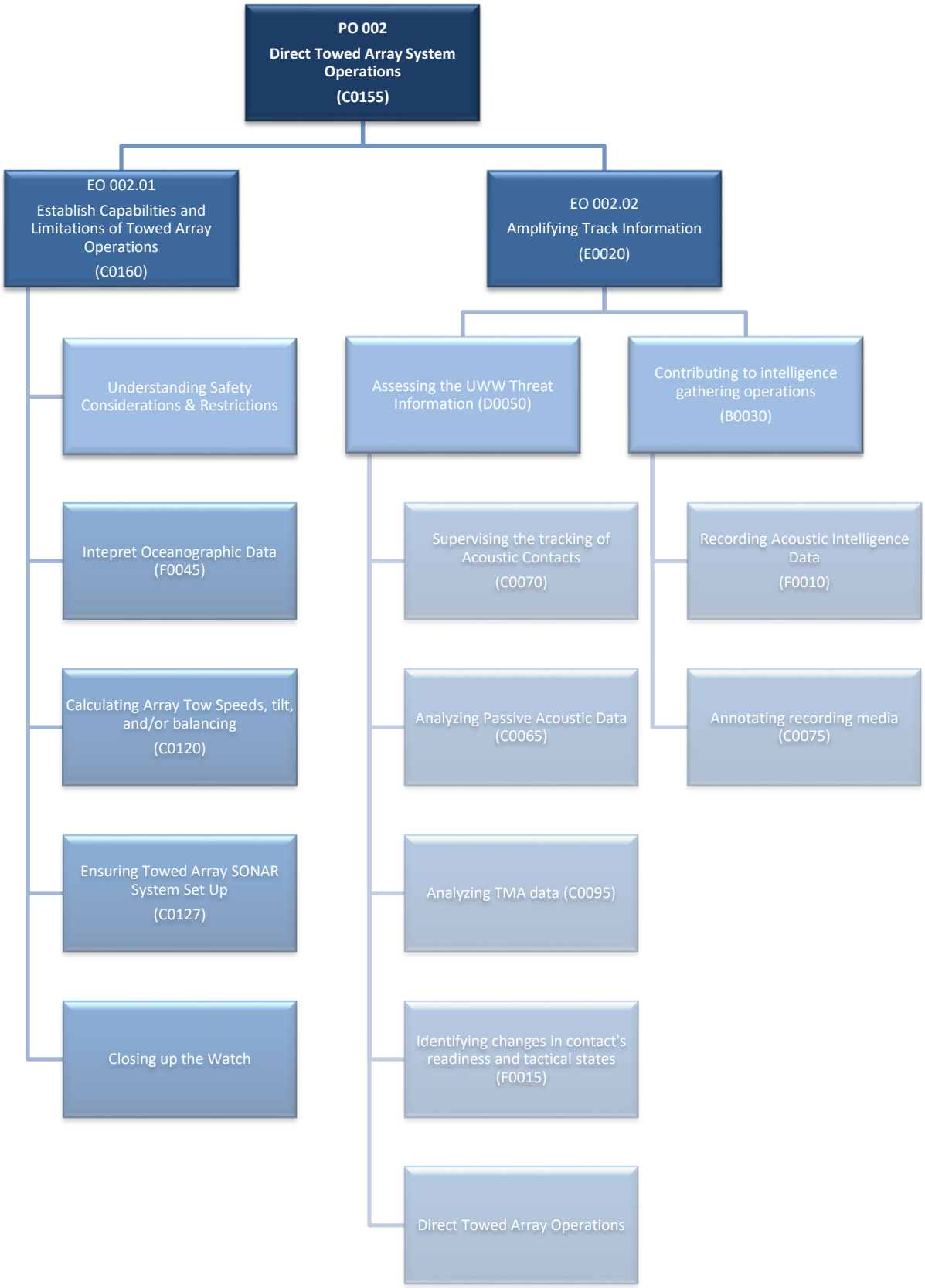
13 Apr – 13 May 2022

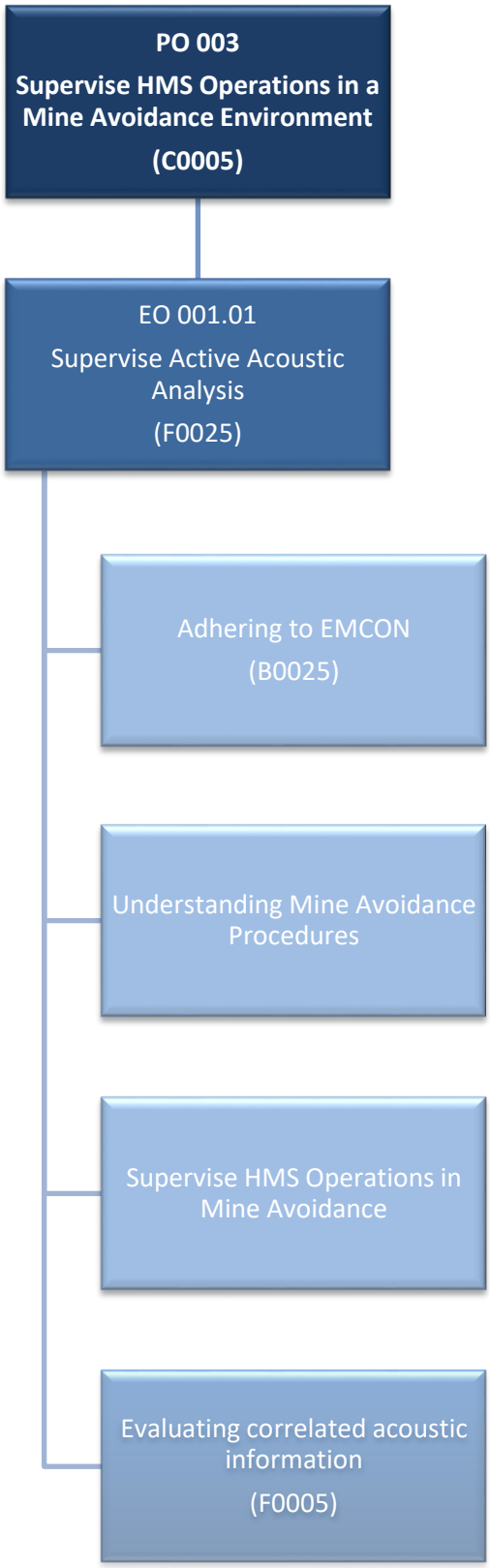
- Conducted final amendments, updates to Annexes and conduct the QSP TDO Checklist review.

ANNEX D – TASK ANALYSIS DIAGRAM

This diagram represents the recreation or original task analysis (scalar) used to define the job.




PO 002
Direct Towed Array System Operations
(C0155)
EO 002.01
Establish Capabilities and Limitations of Towed Array Operations
(C0160)
EO 002.02
Amplifying Track Information
(E0020)
Understanding Safety Considerations & Restrictions
Intepret Oceanographic Data (F0045)
Calculating Array Tow Speeds, tilt, and/or balancing
(C0120)
Ensuring Towed Array SONAR System Set Up
(C0127)
Closing up the Watch
Assessing the UWW Threat Information (D0050)
Contributing to intelligence gathering operations
(B0030)
Supervising the tracking of Acoustic Contacts
(C0070)
Analyzing Passive Acoustic Data (C0065)
Analyzing TMA data (C0095)
Identifying changes in contact's readiness and tactical states (F0015)
Direct Towed Array Operations
Recording Acoustic Intelligence Data
(F0010)
Annotating recording media (C0075)


PO 003
Supervise HMS Operations in a Mine Avoidance Environment
(C0005)
EO 001.01
Supervise Active Acoustic Analysis
(F0025)
Adhering to EMCON
(B0025)
Understanding Mine Avoidance Procedures
Supervise HMS Operations in Mine Avoidance
Evaluating correlated acoustic information
(F0005)

ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	# Trainees + Instructors /	Unit Price	# Days	Initial Course	Each Successive Course	5 Year Total
Airfare (Return)	1	$1,000.00	n/a	$1,000.00	$1,000.00	$10,000.00
R&Q	1	$36.00	25	$900.00	$900.00	$9,000.00
Incidentals	1	$17.50	25	$437.50	$437.50	$4,375.00
Subtotal				$2,337.50	$2,337.50	$23,375.00
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	12	$50.00	1	$600.00	$600.00	$6,000.00
Classroom (square meters)	30	$13.00	25	$9,750.00	$9,750.00	$97,500.00
Computer requirements	12	$1,000.00	1	$12,000.00	$0.00	$12,000.00
IT Equip (Smart board, etc.)	1	$6,000.00	1	$6,000.00	$0.00	$6,000.00
DWAN connection (Setup)	1	$5,000.00	1	$5,000.00	$0.00	$5,000.00
Link Toolkit (on DVD)	15	$1.00	1	$15.00	$15.00	$150.00
Instructional Materials (Printed / Handouts etc.)	15	$30.00	1	$450.00	$450.00	$4,500.00
Subtotal				$33,815.00	$10,815.00	$131,150.00
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
Instructor (PO1) (incremental)	0	$206.00	0	$0.00	$0.00	$0.00
Instructor (Lt(N)) (incremental)	0	$235.00	0	$0.00	$0.00	$0.00
Incremental Civilian Instructor	0	0	0	$0.00	$0.00	$0.00
Subtotal				$0.00	$0.00	$0.00
Total Estimated Costs				$36,152.50	$13,152.50	$154,525.00

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	Plan/Design training exercise	Performance-based exam	18/23 SATs on non-critical items
EC 001.02	EO 001.02	Assist in individual/team training events	Performance-based exam	28/35 SATs on non-critical items
PC 001	PO 001	Conduct Training Events	Performance-based exam	30/42 SATs on non-critical items
EC 002.01	EO 002.01	Establish the capabilities and limitations of towed array operations	Performance-based exam	18/25 SATs on non-critical items
EC 002.02	EO 002.02	Amplify Track Information	Performance-based exam	22/31 SATs on non-critical items
PC 002	PO 002	Direct Towed Array System Operations	Performance-based exam	34/48 SATs on non-critical items
EC 003.01	EO 003.01	Supervise Active Acoustic Analysis	Performance-based exam	13/17 SATs on non-critical items
PC 003	PO 003	Supervise HMS operations in a mine avoidance environment	Performance-based exam	13/17 SATs on non-critical items

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

If a training methodology is determined to be ineffective and/or inefficient, a Learning Support Request (LSR) may be submitted to correct the deficiency. Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
PO 001 – CONDUCT INDIVIDUAL/TEAM TRAINING EVENTS											
LG 001.01.01	200	Classified Classroom	Secret	1	13	0	13	13	0	Yes	CSNI lab
LG 001.01.02	100	Classified Classroom	Secret	1	13	0	13	13	0	Yes	CSNI lab
LG 001.01.03	150	Classified Classroom	Secret	1	13	0	1	13	0	Yes	
LG 001.01.04	50	Classified Classroom	Unclassified	1	13	1	1	13	0	Yes	DWAN or CSNI
LG 001.01.05	150	Classified Classroom	Secret	1	13	1	1	13	0	Yes	DWAN or CSNI
EC 001.01	250	Classified Classroom	Secret	1	13	1	1	13	0	Yes	
LG 001.02.01	400	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
LG 001.02.02	150	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
LG 001.02.03	150	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
LG 001.02.04	200	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
LG 001.02.05	1200	SEACOT	Secret	1	13	0	0	13	0	Yes	SETT/SAFSIM
EC 001.02	350	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
PC 001 Part 1	50	Classified Classroom	Secret	1	13	0	0	13	0	Yes	
PC 001 Part 2	450	SEACOT	Secret	1	13	0	0	0	0	Yes	SETT/SAFSIM
PO 002 – DIRECT TOWED ARRAY SYSTEM OPERATIONS											
LG 002.01.02	200	Unclassified Classroom	Unclassified	1	13	13	0	13	1	Yes	DWAN
LG 002.01.01	50	Unclassified Classroom	Unclassified	1	13	13	0	13	1	Yes	DWAN
LG 002.01.03	100	Unclassified Classroom	Unclassified	1	13	13	0	13	1	Yes	DWAN

LG 002.01.04	150	CMS Lab / SEACOT	Secret	1	0	0	0	0	0	0	CANTASS Mission Simulator or SEACOT
LG 002.01.05	250	CMS Lab / SEACOT	Secret	1	0	0	0	0	0	0	CANTASS Mission Simulator or SEACOT
EC 002.01	400	CMS Lab / SEACOT	Secret	1	0	0	0	0	0	0	CANTASS Mission Simulator or SEACOT
LG 002.02.01	100	Classified Classroom	Secret	1	13	0	0	13	1	Yes	
LG 002.02.04	100	Classified Classroom	Secret	1	13	0	0	13	1	Yes	
LG 002.02.02	200	Unclassified Classroom	Unclassified	1	13	1	0	13	1	Yes	Speakers for video
LG 002.02.03	50	CMS Lab / SEACOT	Secret	1	0	0	0	0	0		CANTASS Mission Simulator or SEACOT
LG 002.02.05	1200	CMS Lab / SEACOT	Secret	1	0	0	0	0	0		CANTASS Mission Simulator or SEACOT
EC 002.02	500	CMS Lab / SEACOT	Secret	1	0	0	0	0	0		CANTASS Mission Simulator or SEACOT
PC 002	600	CMS Lab / SEACOT	Secret	1	0	0	0	0	0		CANTASS Mission Simulator or SEACOT
PO 003 – SUPERVISE HMS OPERATIONS IN A MINE AVOIDANCE ENVIRONMENT											
LG 003.01.01	100	Unclassified Classroom	Unclassified	1	0	1	0	0	0	Yes	
LG 003.01.02	100	Unclassified Classroom	Unclassified	1	0	1	0	0	0	Yes	
LG 003.01.03	600	SEACOT	Secret	1	0	0	0	0	0	Yes	SEACOT
EC 003.01	250	SEACOT	Secret	1	0	0	0	0	0	Yes	SEACOT
PC 003	200	SEACOT	Secret	1	0	0	0	0	0	Yes	SEACOT

RECOMMENDED SCHEDULE

TIME	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Course Introduction & Administration (BAT)**	**PT**	**EC 001.01**– Plan/Design Training Exercise (Plan/Design)	**PT**	**LG 001.02.02** – Create Game Pieces (EDI)
Period 2 0850-0940	**Course Introduction & Administration (TAT)**	**LG 001.01.03** – Compile ASW Tactical Picture (Lecture)	**EC 001.01**– Plan/Design Training Exercise (Plan/Design/Present)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.03** – Create/Input Scenario (EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.01.01** – Interpret Intelligence Data (Lecture)	**LG 001.01.03** – Compile ASW Tactical Picture (Lecture)	**EC 001.01**– Plan/Design Training Exercise (Present)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.03** – Create/Input Scenario (EDI)
Period 4 1040-1130	**LG 001.01.01** – Interpret Intelligence Data (Lecture)	**LG 001.01.03** – Compile ASW Tactical Picture (Lecture)	**EC 001.01**– Plan/Design Training Exercise (Present)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.03** – Create/Input Scenario (EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**LG 001.01.01** – Interpret Intelligence Data (Lecture)	**LG 001.01.04** – Utilizing Tactical Voice Procedures (Activities)	**EC 001.01**– Plan/Design Training Exercise (Present)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.04** – Run Simulator/Trainer System (EDI)
Period 6 1320-1410	**LG 001.01.01** – Interpret Intelligence Data (Lecture)	**LG 001.01.05** – Plan a Training Exercise (Lecture / EDI)	**LG 001.02.01** – Set Up Simulator/Trainer System (Lecture)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.04** – Run Simulator/Trainer System (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.01.02** – Identify Caps & Lims of Platforms (Lecture/Research)	**LG 001.01.05** – Plan a Training Exercise (Lecture / EDI)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.02** – Create Game Pieces (EDI)	**LG 001.02.04** – Run Simulator/Trainer System (EDI)
Period 8 1510-1600	**LG 001.01.02** – Identify Caps & Lims of Platforms (Research)	**LG 001.01.05** – Plan a Training Exercise (Lecture / EDI)	**LG 001.02.01** – Set Up Simulator/Trainer System (EDI)	**LG 001.02.02** – Create Game Pieces (EDI)	**PT**

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**LG 001.02.04** – Run Simulator/Trainer System (EDI)	**PT**	**PT**	**PT**	**EC 001.02**– Assist in Individual/Team Training Events
Period 2 0850-0940	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events
Period 4 1040-1130	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**PC 001 (Part 1)**
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events	**PC 001 (Part 2)**
Period 6 1320-1410	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events	**PC 001 (Part 2)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events	**PC 001 (Part 2)**
Period 8 1510-1600	**LG 001.02.05** – Implement Training Scenarios (EDI)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**LG 001.02.05** – Implement Training Scenarios (Practical)	**EC 001.02**– Assist in Individual/Team Training Events	**PC 001 (Part 2)**

TIME	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**PC 001 (Part 2)**	**LG 002.01.02** – Interpret Oceanographic Data (Lecture)	**PT**	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.04** – Record/Annotate Acoustic Intelligence Data (Lecture)
Period 2 0850-0940	**PC 001 (Part 2)**	**LG 002.01.02** – Interpret Oceanographic Data (Lecture)	**LG 002.01.05** – Closing up the Watch (Practical)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.04** – Record/Annotate Acoustic Intelligence Data (Lecture)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 001 (Part 2)**	**LG 002.01.01** – Safety Considerations and Restrictions (Lecture)	**LG 002.01.05** – Closing up the Watch (Practical)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.02** – Develop TMA (Lecture)
Period 4 1040-1130	**PC 001 (Part 2)**	**LG 002.01.03** – Calculate Array Tow Speeds (Lecture)	**LG 002.01.05** – Closing up the Watch (Practical)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.02** – Develop TMA (Lecture)
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**PC 001 (Part 2)**	**LG 002.01.03** – Calculate Array Tow Speeds (Lecture)	**LG 002.01.05** – Closing up the Watch (Practical)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.02** – Develop TMA (Lecture)
Period 6 1320-1410	**LG 002.01.02** – Interpret Oceanographic Data (Lecture)	**LG 002.01.04** – Towed Array System Set-Up (EDI)	**LG 002.01.05** – Closing up the Watch (Practical)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.02** – Develop TMA (Lecture)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 002.01.02** – Interpret Oceanographic Data (Lecture)	**LG 002.01.04** – Towed Array System Set-Up (EDI)	**LG 002.02.01** – Supervise Tracking Acoustic Contacts (Table Top)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**LG 002.02.03** – Analyzing TMA (EDI)
Period 8 1510-1600	**PT**	**LG 002.01.04** – Towed Array System Set-Up (EDI)	**LG 002.02.01** – Supervise Tracking Acoustic Contacts (Table Top)	**EC 002.01**– Establish Caps & Lims of Towed Array Ops	**PT**

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**LG 002.02.05** – Direct Towed Array Operations (EDI)	**PT**	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information
Period 2 0850-0940	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information	**EC 002.02**– Amplify Track Information
Period 4 1040-1130	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information	**EC 002.02**– Amplify Track Information
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information	**LG 003.01.01** – Supervise Marine Mammal Mitigation (Student Reading)
Period 6 1320-1410	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information	**LG 003.01.01** – Supervise Marine Mammal Mitigation (Student Reading)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**EC 002.02**– Amplify Track Information	**LG 003.01.02** – Understand Mine Avoidance Procedures (Group Discussion)
Period 8 1510-1600	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**LG 002.02.05** – Direct Towed Array Operations (Practical)	**PT**	**EC 002.02**– Amplify Track Information	**PT**

TIME	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**PC 002**	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (EDI)	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**PC 003**
Period 2 0850-0940	**PC 002**	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (EDI/Practical)	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**PC 003**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 002**	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**EC 003.01**– Supervise Active Acoustic Analysis	**PC 003**
Period 4 1040-1130	**PC 002**	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**EC 003.01**– Supervise Active Acoustic Analysis	**PC 003**
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**PC 002**	**PT**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**EC 003.01**– Supervise Active Acoustic Analysis	**PT**
Period 6 1320-1410	**PC 002**	**LG 003.01.02** – Understand Mine Avoidance Procedures (Group Discussion)	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**EC 003.01**– Supervise Active Acoustic Analysis	**BAT (Graduation)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (EDI)	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**EC 003.01**– Supervise Active Acoustic Analysis	**TAT (Course Critique /Reports)**
Period 8 1510-1600	**PC 002**	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (EDI)	**LG 003.01.03** – Supervise HMS Operations in Mine Avoidance (Practical)	**PT**	**Out Routine**